

Constraint-Aware Multilevel Picard Methods

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Abstract

Multilevel Picard (MLP) methods repeatedly feed finite-sample value–gradient estimates into a nonlinear generator. We study a simple failure mode: a noisy child state can leave a region that PDE structure certifies for the exact solution, and the generator can amplify that infeasible component before Monte Carlo averaging has a chance to cancel it. We propose constraint-aware MLP, which retracts child states into certified feasible sets immediately before nonlinear reuse, with both samplewise nearest-point and batch-coupled gauge realizations. On a known high-dimensional MLP counterexample, a certified active subspace makes the offending generator vanish; recursively enforcing that subspace removes every nonlinear correction pathwise and yields exact full-state MSE $(4 - 2/\pi)/N$, independent of ambient dimension. In a paired 100–160D HJB study, Batch-IR reduces value error by 57–62% relative to Samplewise IR, with gains also persisting in deeper nonlinear-funding recursions. Matched generator and level-correction diagnostics, thresholded-driver ties, and inactive-constraint controls delimit this finite-budget advantage rather than establish universal dominance.

1. Introduction

High-dimensional MLP methods combine Picard iteration with Monte Carlo and multilevel differences to solve semilinear parabolic PDEs without spatial grids (E et al., 2021; Hutzenthaler and Kruse, 2020; Neufeld and Wu, 2025). Their recursive structure creates a numerical issue that is easy to miss: a finite-sample value–gradient state is itself an input to later nonlinear evaluations. Thus an error that would be harmless under linear averaging can become large after a norm, square, threshold, or other nonlinear transformation.

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Suppose PDE structure certifies that the exact state $Y^* = (u, z)$ lies in a set $\mathcal{C}(t, x)$, but a Monte Carlo child $\hat{Y}$ lies outside. The usual recursion evaluates $f(t, x, \hat{Y})$ anyway. Our proposal is minimal:

$$\hat{Y} \longrightarrow \tilde{Y} = \mathcal{R}(\hat{Y}) \in \mathcal{C}(t, x) \longrightarrow f(t, x, \tilde{Y}), \quad (1)$$

where the correction occurs *inside* the tree, before nonlinear reuse. We call this constraint-aware MLP.

We study two realizations. **Samplewise IR-MLP** applies the metric projection independently to each child. **Batch-IR** uses one common gauge factor for a sibling Monte Carlo batch. Samplewise projection has cleaner geometry; batchwise contraction couples samples and moves the empirical mean, but can suppress batch spread more aggressively.

Clipping inside MLP is already used in truncated MLP and SCaSML (Beck et al., 2020; Fan et al., 2026). Our contribution is not the projection primitive or its insertion before a driver call. It is the connection between *PDE-certified value–gradient geometry*, the nonlinear feedback it removes, and the statistical choice of a separable or batch-coupled correction (Section 2).

The main theoretical result is an exact structural rescue of the dimensionality counterexample of Hutzenthaler and Nguyen (2025). The true gradient lies in $\text{span}(e_1)$, on which the offending driver vanishes. Enforcing this structure recursively removes every nonlinear correction pathwise. Ordinary samplewise convergence is a supporting inheritance result, conditional on the assumptions of existing MLP theory, rather than a separate contribution (Appendix A).

Contributions. (1) **Geometry-aware recursive correction.** We formulate a common interface for correcting completed value–gradient child estimates before nonlinear reuse, with samplewise metric projection and a shared batch gauge as two realizations. (2) **Exact structural rescue.** We give a generator-compatible sufficient condition for pathwise collapse on a published counterexample and derive exact dimension-independent MSE for recursive subspace projection; final-only and generator-only ablations isolate the required insertion points. (3) **Mechanism-resolving evidence.** Paired high-dimensional HJB, deeper funding, and gradient-dependent experiments connect final error to generator MSE and the variance of the averaged recursive correction. Thresholded-driver ties, batchwise structural checks,

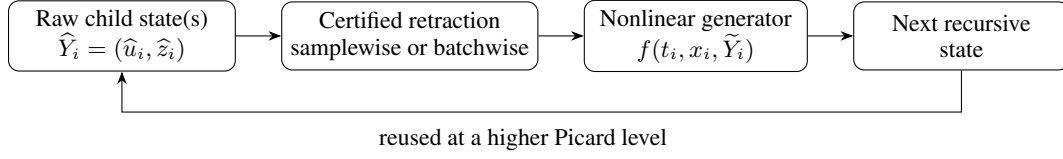


Figure 1. Retract before nonlinear reuse. The correction acts inside the stochastic tree, between each child estimate and the nonlinear generator. Samplewise IR corrects children independently; Batch-IR uses a common feasible scale. A correction only at the final output cannot undo earlier generator evaluations.

and inactive-constraint controls delimit the finite-budget advantage.

2. Related Work

MLP and gradient-dependent solvers. MLP combines Picard iteration, Monte Carlo sampling, and multilevel differences (E et al., 2021). Gradient-dependent variants approximate the solution and its gradient through stochastic representations (Hutzenthaler and Kruse, 2020; Hutzenthaler et al., 2022). Neufeld and Wu (2025) develop a scheme for nonconstant coefficients, and Neufeld et al. (2025) establish polynomial complexity under suitable dimension-dependent assumptions. These works supply our numerical engine and conditional convergence theory; we study the feasible-state corrections inserted within that engine.

Truncated MLP: the closest algorithmic precedent. Beck et al. (2020) use a priori solution bounds to treat locally Lipschitz coercive scalar reactions, including Allen–Cahn equations. Their truncated driver is $f \circ P_r$, with P_r the interval projector; nonexpansiveness transfers a Lipschitz bound to the modified driver (Beck et al., 2020, Eq. (76), Lemma 3.2). Thus PDE-based truncation before nonlinear evaluation and the modified-driver inheritance argument are existing ideas. We extend the numerical focus from scalar reaction truncation to structured value–gradient constraints, including inactive subspaces, joint norm bounds, and normalized sensitivity regions. The substantive questions are which geometry eliminates a harmful gradient-feedback mechanism and how separable versus batch-coupled retractions affect finite-budget error.

Surrogate correction and clipping. SCaSML uses a learned PDE surrogate and solves a semilinear defect equation with an MLP corrector at inference time (Fan et al., 2026). Its experimental implementations also clip estimated values and gradients; clipping is therefore not a distinction by itself. Our intervention concerns the geometry of child states passed to the nonlinear driver, rather than the construction or learning of a defect equation. Surrogate correction reduces the defect to be estimated; our retractions control which noisy states the driver reuses. The mechanisms are

complementary, and our standalone method requires no trained surrogate.

Hard constraints in learned predictors. HardNet embeds hard-constraint enforcement into neural networks and proves universal approximation guarantees for constrained predictors (Min and Azizan, 2025). We share the use of feasible-set geometry, but apply it to recursive Monte Carlo child estimates rather than trainable network outputs. The relevant guarantees concern recursive numerical error rather than approximation capacity of a learned function.

Constraints on the physical process. Zhong (2025) study MLP for drift-control problems with state constraints using reflected dynamics. Those constraints restrict the controlled spatial process. Our sets instead restrict the estimated solution–gradient pair (u, z) at a sampled point, without imposing reflection on the simulated diffusion. The constrained objects, and therefore the numerical modifications, are different.

Dimensional failure and structural rescue. Hutzenthaler and Nguyen (2025) identify an HJB family for which the original full-history recursion suffers dimensional deterioration despite a low-dimensional solution. Building on that failure, we identify a sufficient repair condition: every corrected gradient lies in a certified set on which the offending generator vanishes. This yields pathwise collapse for any compatible retraction and an exact MSE for the subspace projector (Theorem 2). The certificate is supplied by PDE symmetry, not learned from data; after collapse, the estimator equals reduced Monte Carlo. This identifies and repairs an obstruction in the stochastic recurrence while retaining the reduced-dimensional benchmark as the appropriate reference.

3. Constraint-Aware MLP

Consider a semilinear parabolic PDE

$$\partial_t u + \mathcal{L}u + f(t, x, u, z) = 0, \quad u(T, x) = g(x), \quad (2)$$

with gradient-like state $z = \sigma(t, x)^\top \nabla_x u$. MLP recursively estimates $Y = (u, z)$ at random space–time points and reuses those estimates inside f . The relevant failure mode is

Table 1. Same feasibility principle, different statistical geometry.

	Samplewise	Batchwise
map	$\Pi_C Y_i$	$\alpha_B Y_i$
coupling	none	shared
nearest point	yes	no
theory	projection	structural rescue
risk	proj. bias	coupling/bias

not merely that $\hat{Y}$ has large variance; it is that the nonlinear map can destroy cancellation. For example, if the exact gradient obeys $\|z\| \leq 4$ but a finite-sample estimate has norm 12, then the quadratic HJB driver $f(z) = -\frac{1}{2}\|z\|^2$ turns a factor-three state error into a factor-nine driver magnitude. Once this value enters a higher Picard level, a final-output correction is too late. Figure 1 shows the insertion point.

3.1. Two recursive retractions

Let $Y_1, \dots, Y_B$ be completed sibling child-state estimates at sampled points (t_i, x_i) , with certified sets $\mathcal{C}_i = \mathcal{C}(t_i, x_i)$. The formulas suppress the index when all sets coincide.

Samplewise IR. For nonempty closed convex $\mathcal{C}$,

$$\tilde{Y}_i = \Pi_{\mathcal{C}} Y_i. \quad (3)$$

This is local, introduces no cross-sample coupling, and makes the smallest Euclidean correction to each child.

Batch-IR. For a gauge set $\mathcal{C} = \{y : \gamma_{\mathcal{C}}(y) \leq 1\}$, define

$$\alpha_B = \min \left\{ 1, \frac{1}{\max_i \gamma_{\mathcal{C}_i}(Y_i)} \right\}, \quad \tilde{Y}_i = \alpha_B Y_i. \quad (4)$$

For a common ball, $\alpha_B = \min\{1, R/\max_i \|Y_i\|\}$. At distinct sampled points each child uses its own gauge. The gradient-only experiments leave the value component unchanged. A common scale preserves relative directions and magnitudes in the constrained coordinates, but couples siblings and moves their empirical mean.

3.2. Target Preservation and Local Work

The key deterministic geometry is the projection inequality: for every feasible $Y^* \in \mathcal{C}$,

$$\|\Pi_{\mathcal{C}} Y - Y^*\|^2 \leq \|Y - Y^*\|^2 - \text{dist}(Y, \mathcal{C})^2. \quad (5)$$

Thus samplewise projection removes at least the squared overshoot component of child-state error.

Proposition 1 (Target preservation). *If $Y^*(t, x) \in \mathcal{C}(t, x)$ and the retraction is the identity on feasible states, then constraint-aware evaluation leaves the exact generator unchanged:*

$$f(t, x, \mathcal{R}(Y^*(t, x))) = f(t, x, Y^*(t, x)).$$

Hence the correction stabilizes off-target stochastic states rather than changing the intended PDE solution.

Proof. Feasibility gives $\mathcal{R}(Y^*) = Y^*$. $\square$

For samplewise convex projection, this can be strengthened under the standard assumptions used by gradient-dependent MLP theory: defining $\bar{f}(t, x, y) = f(t, x, \Pi_{\mathcal{C}(t, x)} y)$ yields an ordinary MLP problem with the same feasible solution, and nonexpansiveness of projection does not increase the state Lipschitz constant. We give the short inheritance argument in Appendix A.

Work order. Projection onto the coordinate subspaces, balls, and diagonal-normalized ellipsoids used below costs $O(d)$ per state. Forming a batch gauge and rescaling B d -dimensional states costs $O(Bd)$, the same arithmetic order as reading the batch. When the projected problem satisfies the cited convergence theorem with the same parameter scalings, this local overhead preserves its polynomial complexity exponents. For Batch-IR we claim only unchanged per-update arithmetic order; a sharp cost-to-accuracy theory must account for coupling and is future work.

4. Where Do the Constraints Come From?

The method is useful only when $\mathcal{C}(t, x)$ is certified independently of the noisy solver output. We use three kinds of structure. First, translation symmetry can force inactive gradient coordinates to be exactly zero, producing a linear subspace. Second, Hopf–Cole or Lipschitz arguments can give norm balls for gradients. Third, financial sensitivities can yield state-dependent ellipsoids in normalized coordinates. These are not thresholds tuned to realized test error.

For the Rosenbrock HJB benchmark, the terminal condition is $g(x) = \log((1 + x^\top A x)/2)$ and $\|\nabla g(x)\| \leq \sqrt{\lambda_{\max}(A)}$. The Hopf–Cole representation carries this envelope backward; with the benchmark convention $z = \sqrt{2}\nabla u$ and $\lambda_{\max}(A) \leq 7.5$, we obtain the non-oracle ball $\|z\| \leq \sqrt{15}$. In the Neufeld–Wu benchmark, a flow-sensitivity bound places the exact gradient well below the threshold where the driver activates. In the funding problem, the certificate is an ellipsoid in normalized delta coordinates. Derivations are collected in Appendix C.

5. Exact Structural Rescue

Consider the family studied by Hutzenthaler and Nguyen (2025):

$$g_d(x) = |x_1|, \quad f_d(v) = \left(\sum_{j=2}^d v_j^2 \right)^{1/2}, \quad (6)$$

$$\partial_t u_d + \frac{1}{2} \Delta u_d + f_d(\nabla u_d) = 0, \quad u_d(1, x) = g_d(x). \quad (7)$$

Although the target depends only on x_1 , the full-history recursion estimates all d gradient coordinates. Noise in coordinates $2:d$ is nonnegative after f_d and becomes a spurious recursive source. The source paper proves that the unprojected method suffers a dimensional lower bound with depth-dependent quantifiers; Appendix B states it precisely.

Translation symmetry certifies the active subspace

$$S_d = \text{span}(e_1), \quad \nabla u_d(t, x) \in S_d,$$

and the driver is exactly compatible with it:

$$f_d(v) = 0 \quad \text{for all } v \in S_d. \quad (8)$$

Theorem 2 (Structural rescue and dimension-free error). *Apply any recursive correction before every f_d evaluation such that each corrected child gradient lies in S_d almost surely. Then all nonlinear multilevel corrections vanish pathwise. In particular, for orthogonal projection Q_d onto S_d at every recursive return, including the final gradient, at $(t, x) = (0, 0)$ with $N = m^n$ terminal samples,*

$$\mathbb{E} \left\| \widehat{Y}_{n,m}^{\text{IR}} - Y^* \right\|^2 = \frac{4 - 2/\pi}{N}, \quad (9)$$

independent of ambient dimension d . Any batch-coupled correction of the form $v_i \mapsto \alpha_B Q_d v_i$, $0 \leq \alpha_B \leq 1$, has the same pathwise nonlinear collapse.

Proof. Equation (8) makes every corrected generator value zero, recursively, so the full nonlinear tree collapses to the projected terminal block. At the origin the only nonzero estimated components are $N^{-1} \sum_i |G_i|$ and $N^{-1} \sum_i |G_i| G_i$, with $G_i \sim \mathcal{N}(0, 1)$. They are unbiased with variances $(1 - 2/\pi)/N$ and $3/N$; summing gives (9). The batch statement follows because $\alpha_B Q_d v_i \in S_d$. $\square$

The theorem identifies the essential property: not clipping magnitude, but returning to a *generator-compatible certified set*. Final-only projection cannot undo earlier nonlinear feedback; generator-only projection fixes the value path but leaves inactive gradient noise in the reported state. Exact ablation identities are in Appendix B.

6. Experiments

6.1. Counterexample: the mechanism is visible

We reproduce the source recurrence using its random-time law $\mathbb{P}(r \leq b) = \sqrt{b}$, independent child trees in each multilevel difference, and matched random trees across methods. We test $d \in \{2, 5, 10, 20, 50, 100, 200, 500, 1000\}$ and $n = m \in \{1, 2, 3, 4\}$ with 512 estimator replicas per configuration. Time and active-coordinate streams are

coupled across dimensions, so a flat subspace-IR curve is an expected consequence of the theorem rather than post-hoc smoothing. We separately validate the closed-form MSE with 65,536 fresh Gaussian replicas for each $N \in \{1, 4, 27, 256\}$.

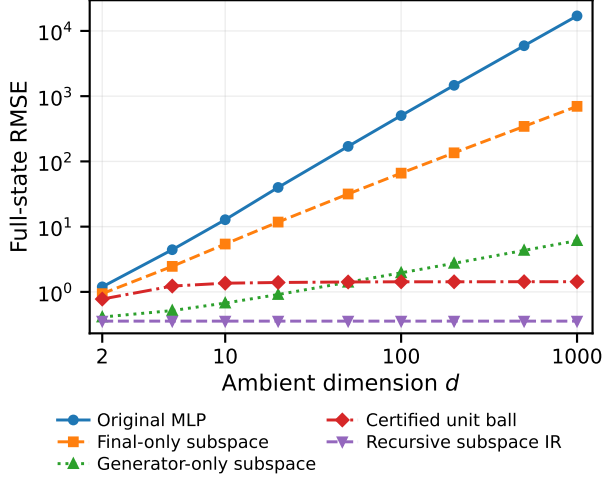
Figure 2 separates full-state error from value error (selected values are in Table 2). At $d = 1000$, raw MLP and final-only projection have the same value RMSE, 396.978, whereas recursive subspace IR has value RMSE 0.119. Thus the gain is not an artifact of deleting gradient coordinates from the final metric. The exact MSE prediction in (9) agrees with an independent Gaussian-moment validation to within 0.56% across $N \in \{1, 4, 27, 256\}$. Appendix B keeps the independent MSE check, the separate primary-sweep RMSE comparison, and the full depth sweep (Figures A1–A2).

6.2. Headline HJB and Deeper Funding

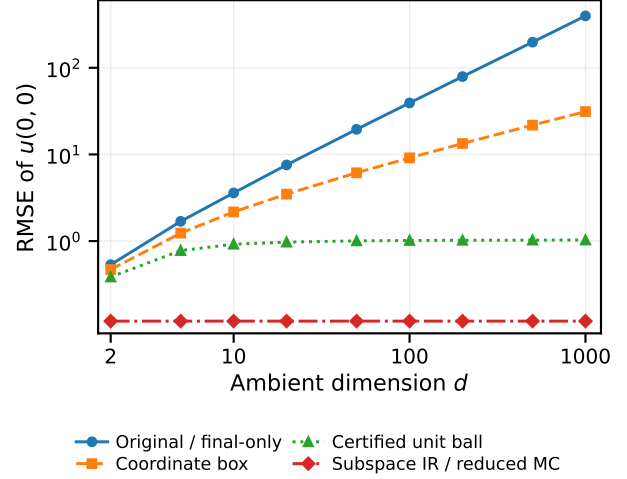
We extend the paired study to the headline Rosenbrock HJB protocol and deeper funding recursions. Within each configuration the three methods share random draws, and only child gradients are corrected before driver evaluation; the final value is never clipped. These are standalone MLP experiments, not comparisons against a trained SCaSMML pipeline. Earlier geometry ablations, diagnostics, and mean-preserving controls are retained in Appendix D; the extension’s protocols and complete tables are in Appendix E.

100–160D Rosenbrock HJB. We use $n = 2$, $M = 10$, a fixed matrix at each dimension, 1,000 interior unit-ball points and 200 unit-sphere points with sampled times, and 10 paired repetitions. The radius is the non-oracle $\sqrt{15}$; the reference uses scaled Hopf–Cole quadrature. Figure 3a shows Samplewise IR errors of 0.621–0.640 and Batch-IR errors of 0.243–0.265: a 57.3–62.1% reduction, with Batch-IR lower in all 10 repetitions at each dimension. The uncorrected errors are 2.280–3.479. These raw values must not be substituted for the older heuristic-clipping results in Figure A3(a).

Deeper 100D funding. We use the same normalized-delta region for both corrections, with 100 paired root estimates per setting. The three depth/budget pairs in Figure 3b have Samplewise/Batch-IR MAEs of 0.5215/0.4566, 0.3839/0.3001, and 0.7708/0.6897, respectively. The gain therefore persists at depths three and four; it is not confined to the earlier shallow sweep. The corresponding reductions are 12.4%, 21.8%, and 10.5%, with lower absolute error on 73%, 75%, and 60% of paired estimates. Different budgets preclude a monotone-depth or wall-clock claim. The earlier funding work envelope remains in Figure A5.

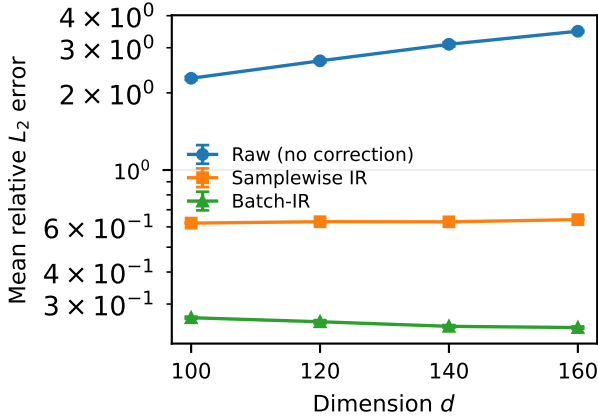
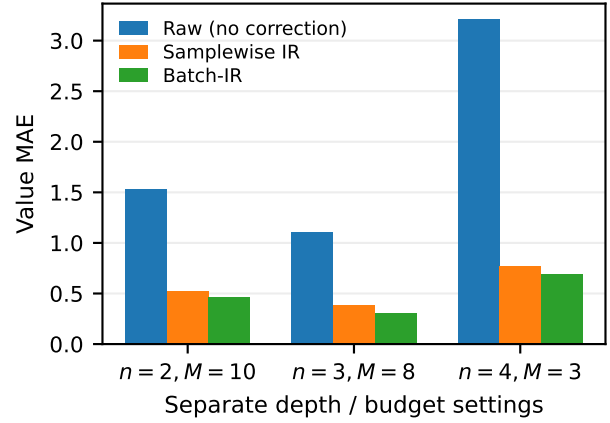


(a) Full value-gradient state.



(b) Value error alone.

Figure 2. Recursive feasibility repairs more than the final gradient metric. All nine dimensions are shown at $n = m = 3$, with 512 paired replicas. (a) Full-state error distinguishes final-only, generator-only, norm-ball, and recursive subspace corrections. (b) Raw MLP and final-only correction have identical value outputs; recursive subspace IR, generator-only projection, and reduced MC coincide in value. Shared active-coordinate streams explain the flat IR curve. These are finite-depth comparisons, not a fitted asymptotic lower bound.


 (a) Rosenbrock HJB: $n = 2$, $M = 10$.


(b) 100D funding: three depth/budget settings.

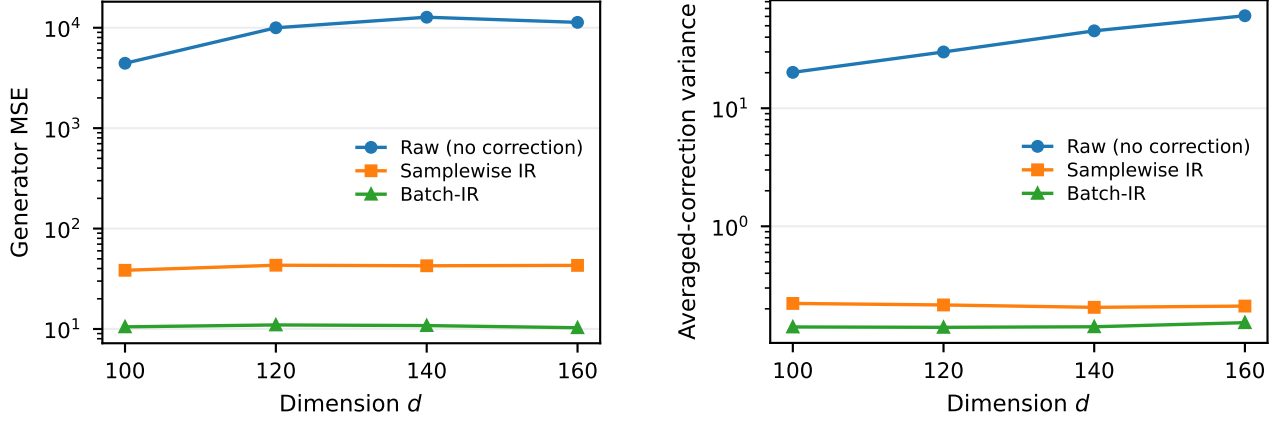
Figure 3. Batch-IR gains extend to headline dimensions and deeper recursion. (a) Relative L_2 value error over 1,200 fixed space-time points at each dimension; error bars are one sample standard deviation across 10 paired repetitions, often smaller than the markers. Raw means no correction, not the older heuristic baseline. (b) MAE over 100 paired root estimates per setting, against the reference value 21.299. Both n and M change, so this is not a fixed-work depth comparison. Different metrics are compared only within each panel.

6.3. Matched Mechanism Diagnostics

On the same HJB study, generator MSE falls from 38.42–43.17 for Samplewise IR to 10.30–10.98 for Batch-IR. The variance across repetitions of the *actual averaged level correction* falls from 0.206–0.222 to 0.139–0.153 (Figure 4). Relative to raw MLP, the respective reductions are 99.76–99.92% and 99.30–99.75%.

The distinction matters: the pooled variance of individual generator values is *not* uniformly reduced. At three of four

dimensions it rises relative to Samplewise IR even though the averaged correction has lower variance. Batch-IR couples siblings, so individual dispersion alone does not determine correction variance. Its common scale activates for 99.87–99.98% of HJB batches, with mean scale decreasing from 0.330 to 0.260. These results show an aggressive finite-budget correction, not a vanishing-bias or consistency theorem. The earlier two-method diagnostics are preserved in Figure A4.



(a) Generator MSE at matched child points.

(b) Variance of the averaged nonlinear correction.

Figure 4. The batchwise extension improves two distinct recursive diagnostics. (a) Generator MSE is averaged over 6,400 sampled child states, evaluated with a stable reference gradient. (b) At each of the 1,200 fixed parent points, variance is computed across 10 repetitions and then averaged over points. This includes the effect of sibling coupling. Pooled individual-increment variance is a different statistic: it is *higher* for Batch-IR than Samplewise IR at $d = 120, 140, 160$ (Table 8).

6.4. Overshoot and an Independent Positive Benchmark

The earlier mechanism and benchmark evidence answers questions beyond whether Batch-IR beats Samplewise IR. Figure 5(a) reproduces the 100D, $n = 2$ overshoot sweep over six sampling bases and six valid radius factors, correcting only child gradients and leaving the final value untouched. The descriptive Spearman correlations are 0.886 for overshoot energy versus final gain and 0.995 for violation frequency versus gain. Larger radii and budgets share data; this is not an independent-sample causal test. Complete new draws and corrected-reference details are in Appendix F.

The original Neufeld–Wu experiment in Figure 5b shows gains over raw MLP at all six dimension–budget pairs (Table 4). This is distinct from comparing two feasible corrections. The older HJB geometry comparison remains in Figure A3(a), and the funding budget envelope remains in Figure A5. A recovered funding ablation reaches $n = 5$ and retains the coordinate-box baseline (Appendix G); it is not replaced by the new three-method study.

6.5. Ties, Structural Checks, and Negative Controls

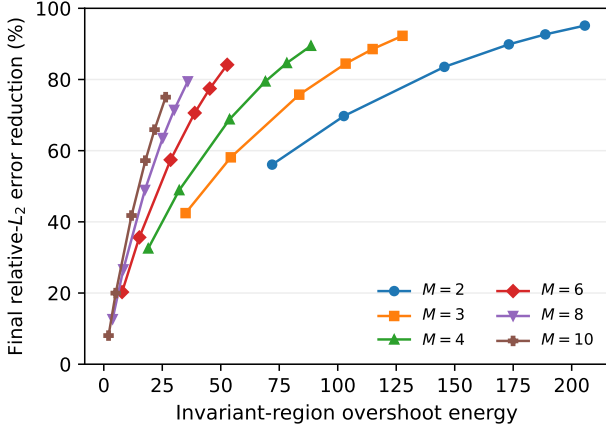
Thresholded Neufeld–Wu driver. At $n = M = 2$, 30 runs at each of $d = 100, 200, 300$ give Samplewise IR and Batch-IR identical MAEs of 0.02131, 0.01682, and 0.01737, versus raw errors 0.06066, 0.03878, and 0.04039. Both corrections keep the driver inactive, as in the earlier $n = M = 3$ comparison. These are ties, not batchwise wins (Table 11). The implementation retains a finite-step forward approximation, so these are not exact-time convergence measurements.

Batchwise structural sanity check. A separate 512-replica study at $n = m = 3$ tests $z_i \mapsto \alpha_B Q_d z_i$ for $d = 10, 100, 1000$. Corrected *value* outputs agree realization by realization, with maximum discrepancy zero and value RMSE 0.110083. The batch scale activates in 58.98% of tested groups. Equality therefore reflects generator compatibility, not an inactive correction. The batch implementation does not project its returned root gradient; this test is not a verification of equal full-state MSE (Appendix E.5).

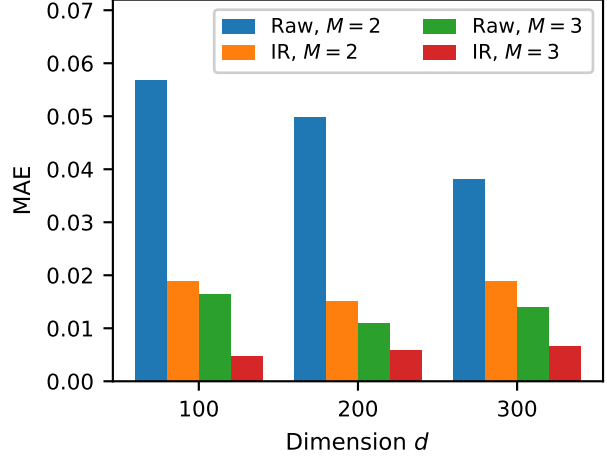
Negative controls and mean preservation. The repository’s paired Batch-IR controls confirm exact value-output overlap in Allen–Cahn and credit risk when constraints are inactive. In linear convection–diffusion, batch correction activates in 100%, 97%, and 8.5% of recorded groups at the three budgets, yet the value remains unchanged because $f = 0$ (Appendix I). The earlier panels and mean-preserving comparisons remain in Figures A8 and A6. A controlled surrogate-quality study, including cases where IR loses to raw defect MLP, is retained in Appendix H; it is not a trained-SCaSML comparison.

7. What the Experiments Establish

The counterexample separates a structural claim from a statistical design choice. The structural claim is that recursive infeasibility can create a spurious nonlinear source, and that a generator-compatible certificate can remove it exactly. The statistical choice is how to move noisy samples into the feasible set. Equation (5) favors nearest-point projection for one-state Euclidean error; Figures 3 and A6 show that a common batch scale can nevertheless be better after nonlinear recursion. This is not a contradiction: two complete



(a) 36-setting HJB overshoot/gain reproduction.



(b) Independent Neufeld–Wu multi-budget results.

Figure 5. Mechanism evidence and a positive PDE benchmark are complementary. (a) A newly executed reproduction of the earlier overshoot diagnostic, using scaled reference quadrature: each curve fixes M and varies the valid radius. It is not the original random sample or a digitized plot. (b) The original two-budget, 30-run Neufeld–Wu result is retained unchanged, with 47.5–71.4% MAE reduction over raw MLP. Samplewise/Batch-IR ties in later experiments do not erase this positive comparison against raw MLP.

recursive schemes generate different later states, and batch coupling changes the mean, dispersion, and cross-sibling dependence. The new diagnostics support lower generator MSE and averaged-correction variance, not uniform reduction of every variance statistic.

A mean-preserving batch diagnostic sharpens the point. When the raw batch mean is feasible, shrinking deviations around that mean can improve final error without changing the empirical mean. When the batch mean itself is outside a convex feasible set, no all-feasible mean-preserving correction exists, because the mean of feasible points must be feasible. Thus low-budget high-dimensional failure can affect the batch center as well as isolated outliers.

8. Limitations

The exact rescue uses known PDE structure rather than discovering an active subspace, and its collapsed estimator equals reduced Monte Carlo. Approximate or learned regions need a separate misspecification analysis. Feasibility preserves the intended solution, but does not alone prove consistency or order the final errors of two different recursive solvers. Samplewise inheritance still requires the regularity, moment, norm, and dimension-scaling assumptions of the cited MLP theorem. Batch-IR changes dependence and finite-sample bias; its sharp consistency and cost-to-accuracy theory remain future work. The finance envelope also requires the additional justification noted in Appendix C. The extension uses 10 HJB repetitions on fixed test sets and matrices, 100 funding estimates against a numerical reference, and finite-step Neufeld–Wu dynam-

ics. These finite-budget comparisons do not prove universal dominance over MLP or trained surrogate solvers.

9. Conclusion

MLP can amplify Monte Carlo noise when infeasible intermediate states are passed through nonlinear generators. Constraint-aware MLP inserts a certified retraction exactly at that interface. The strongest result is structural: on a published dimensionality counterexample, enforcing the PDE-certified active subspace removes all nonlinear corrections pathwise and changes the error from ambient-dimension dependent to the exact dimension-free rate $(4 - 2/\pi)/N$. Beyond that example, paired 100–160D HJB and deeper funding experiments favor Batch-IR, while thresholded-driver ties and inactive-constraint controls delimit the gain. Matched diagnostics isolate lower generator MSE and averaged-correction variance without claiming lower individual-increment variance everywhere. The central contribution is the link between certified value–gradient geometry and the nonlinear feedback it removes, supported by an exact rescue theorem and paired evidence on separable versus batch-coupled corrections.

Impact Statement

This work develops stochastic PDE solvers. Its broader impacts arise through scientific and financial simulation; no application-specific societal impact is singled out here.

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A. Samplewise Convergence Inheritance

This appendix records a conditional value–gradient version of the modified-driver argument already used for scalar truncated MLP by Beck et al. (2020, Eq. (76), Lemma 3.2). It supplies a correctness check for the specified insertion point, not a new full-history convergence analysis.

For each (t, x) let $\mathcal{C}(t, x)$ be nonempty, closed, and convex, let $\Pi_{t,x}$ be its metric projector, and define

$$\bar{f}(t, x, y) = f(t, x, \Pi_{t,x}y). \quad (10)$$

Assume $(t, x, y) \mapsto \Pi_{t,x}y$ is measurable, the exact state $Y^*(t, x)$ is feasible, f is uniformly L -Lipschitz on $\mathcal{C}(t, x)$ in its state argument, and the remaining coefficient, terminal-data, moment, and random-time assumptions of the chosen gradient-dependent MLP theorem hold for $\bar{f}$.

Proposition 3 (Inheritance for pointwise Samplewise IR). *Under these assumptions, $\bar{f}$ is globally L -Lipschitz in y , the intended feasible solution also solves the PDE with f replaced by $\bar{f}$, and the recursion that projects each child only immediately before a driver call is pathwise identical to ordinary MLP applied to $\bar{f}$ on the same random tree. Hence the corresponding published convergence and complexity bounds apply to Samplewise IR under their remaining assumptions.*

Proof. Metric projection is nonexpansive, so

$$\begin{aligned} |\bar{f}(t, x, y) - \bar{f}(t, x, y')| &\leq L \|\Pi_{t,x}y - \Pi_{t,x}y'\| \\ &\leq L \|y - y'\|. \end{aligned}$$

Feasibility gives $\Pi_{t,x}Y^* = Y^*$, hence $\bar{f}(t, x, Y^*) = f(t, x, Y^*)$. Finally, at every finite MLP depth, replacing a child Y by $\Pi_{t,x}Y$ before evaluating f produces exactly the call $\bar{f}(t, x, Y)$. Induction on depth gives pathwise identity with ordinary MLP for $\bar{f}$, after which the existing theorem applies (Hutzenthaler and Kruse, 2020; Hutzenthaler et al., 2022; Neufeld and Wu, 2025; Neufeld et al., 2025). $\square$

This argument does *not* cover Batch-IR because α_B depends jointly on siblings. It also does not cover a different algorithm that first projects samples and then averages them into a new state estimator: for nonlinear Π , generally $\mathbb{E}[\Pi(Y)] \neq \Pi(\mathbb{E}Y)$. Fixed affine/subspace projections are special because they commute with averaging. A diagonal-normalized ellipsoid uses a weighted metric, not necessarily the Euclidean metric; applying an inheritance theorem additionally requires its norm, regularity, and dimension-dependent bounds to be checked in the same coordinates. Euclidean nonexpansiveness alone does not establish the dimension-uniform coordinate or maximum-norm bounds required by a particular complexity theorem.

B. Counterexample Details and Ablations

B.1. Source lower bound and certification

Let $U_{n,m}^d$ denote the source full-history recursion and set $Y_d^* = (u_d, \nabla u_d)$. Hutzenthaler and Nguyen (2025) show that for every $p \geq 0$ and every integer $n > 2p$,

$$\liminf_{d \rightarrow \infty} d^{-p} \left(\mathbb{E} \|U_{n,n}^d(0, 0) - Y_d^*(0, 0)\|^2 \right)^{1/2} = \infty.$$

This is a depth-dependent asymptotic statement, not a claim that one fixed depth grows faster than every polynomial.

For any translation a with $a_1 = 0$, $u_d(t, x + a)$ has the same PDE and terminal data as $u_d(t, x)$. Uniqueness gives invariance in coordinates $2:d$, hence $\partial_{x_j} u_d = 0$ for $j \geq 2$. The PDE reduces to the heat equation and

$$u_d(t, x) = \mathbb{E}|x_1 + \sqrt{1-t}G|, \quad G \sim \mathcal{N}(0, 1).$$

B.2. Ablations

If subspace projection is used only inside the generator but not on the returned gradient, nonlinear corrections still vanish, but the inactive final-gradient coordinates remain. At the origin,

$$\mathbb{E} \left\| \widehat{Y}_{\text{gen-only}} - Y^* \right\|^2 = \frac{d + 3 - 2/\pi}{N}.$$

If projection is applied only at the final state, the value component is unchanged realization by realization from raw MLP, because all earlier generator calls are identical.

For the first terminal block,

$$\mathbb{E} \|V_{\perp, N}\|^2 = \frac{d-1}{N}, \quad \mathbb{E} f_d(V_N)^2 = \frac{d-1}{N},$$

$$f_d(Q_d V_N) = 0.$$

This quantifies the first conversion of centered inactive-gradient noise into a positive nonlinear source.

B.3. Independent Validation and Depth Sweep

Figure A1 deliberately separates two different checks. The left panel uses 65,536 fresh Gaussian replicas per terminal sample count, independently of the full MLP sweep. Its error bars are normal-approximation 95% intervals for empirical MSE. The right panel uses the 512 replicas of the main paired MLP experiment and reports RMSE as a function of depth. They should not be described as the same experiment.

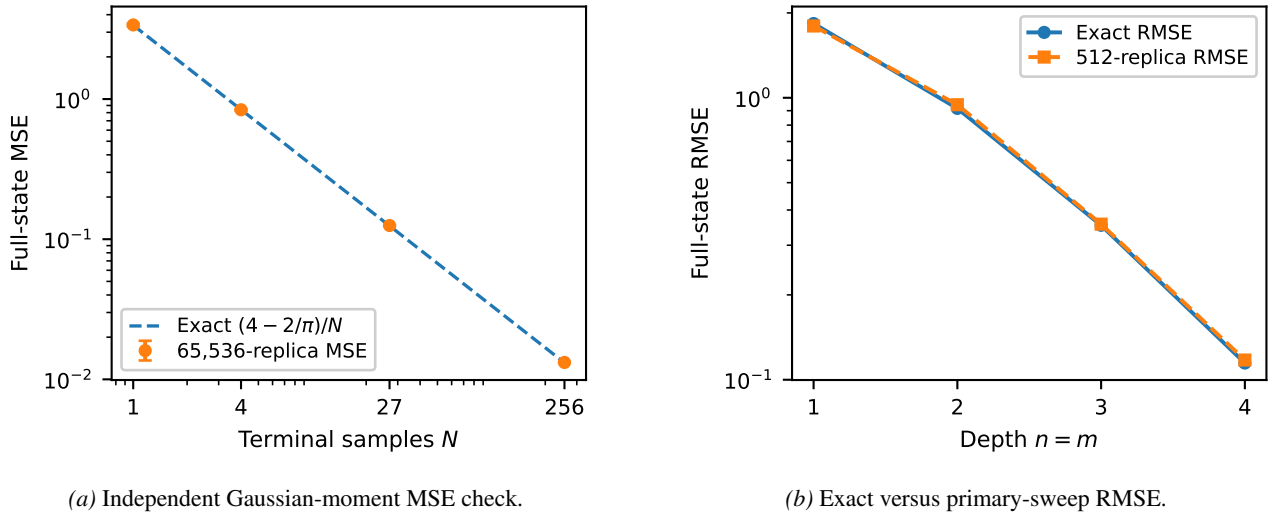


Figure A1. Two validations of the rescue theorem. (a) The empirical MSE differs from $(4 - 2/\pi)/N$ by less than 0.56% at each of $N \in \{1, 4, 27, 256\}$; small intervals reflect the larger replicate count. (b) The original 512-replica RMSE comparison is retained separately. Common active-coordinate streams make this second curve identical across tested ambient dimensions.

Figure A2 retains all four raw-MLP depth curves across all nine tested dimensions. At $d = 1000$, $n = m = 4$, raw full-state RMSE is 239406.128, compared with 0.117464 for recursive subspace IR. The plot is a finite-depth diagnostic. The source lower bound above, rather than an extrapolated fit, supplies the asymptotic statement.

Table 2. Full-state RMSE at $n = m = 3$ ($N = 27$).

d	Raw MLP	Final only	Unit ball	Subspace IR
2	1.196	0.939	0.776	0.356
10	12.767	5.412	1.358	0.356
100	504.759	66.088	1.427	0.356
1000	16995.683	697.972	1.434	0.356

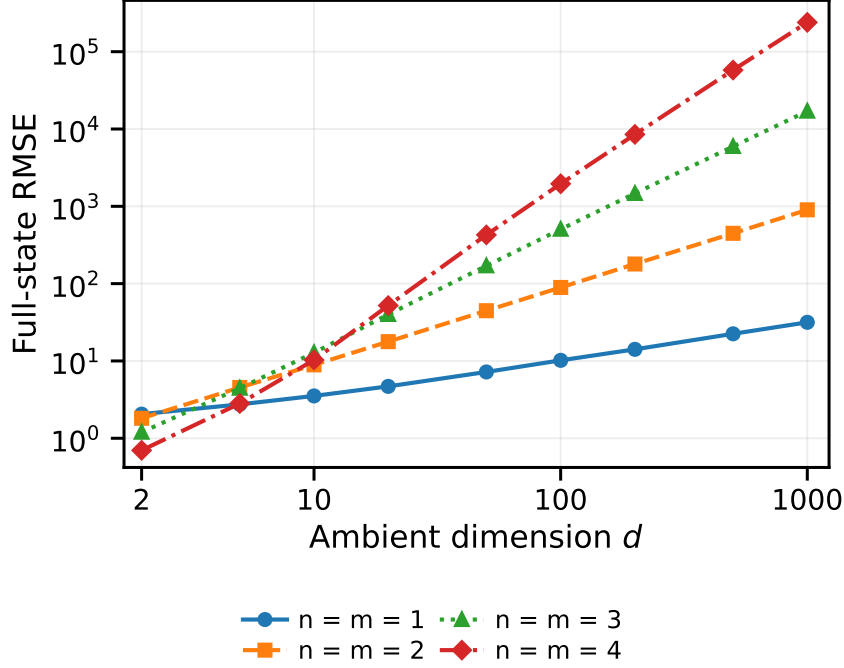


Figure A2. **Full finite-depth counterexample sweep.** Increasing depth steepens the dimensional deterioration of the raw recursion over the tested range. The source theorem quantifies depth as $n > 2p$ for each polynomial exponent p ; this plot does not assert that one fixed depth grows faster than every polynomial. All plotted vertices are retained from the archived vector figure.

C. Certificates Used in the Broader Benchmarks

Rosenbrock HJB. For $u_t + \Delta u - \|\nabla u\|^2 = 0$ and $g(x) = \log((1 + x^\top Ax)/2)$,

$$\nabla g(x) = \frac{2Ax}{1 + x^\top Ax}, \quad \|\nabla g(x)\| \leq \sqrt{\lambda_{\max}(A)}.$$

The Hopf–Cole representation carries the same envelope backward in time. With the benchmark convention $z = \sqrt{2}\nabla u$ and coefficient range giving $\lambda_{\max}(A) \leq 7.5$, we use $\|z\| \leq \sqrt{15}$.

Neufeld–Wu. The public example has linear drift $\mu(x) = 0.06x$, additive diffusion, a 1-Lipschitz terminal payoff, and a thresholded driver proportional to $(\|z\|_\infty - 25)_+$. Flow sensitivity gives $\|\nabla u(t, \cdot)\| \leq e^{0.06(T-t)} \leq e^{0.015}$ over the benchmark horizon, well below the activation threshold.

Nonlinear funding. For the 100D borrowing/lending benchmark, a model-derived delta envelope yields

$$\sum_{i=1}^d \left(\frac{z_i}{\sigma x_i} \right)^2 \leq \exp(\sigma^2(T-t)),$$

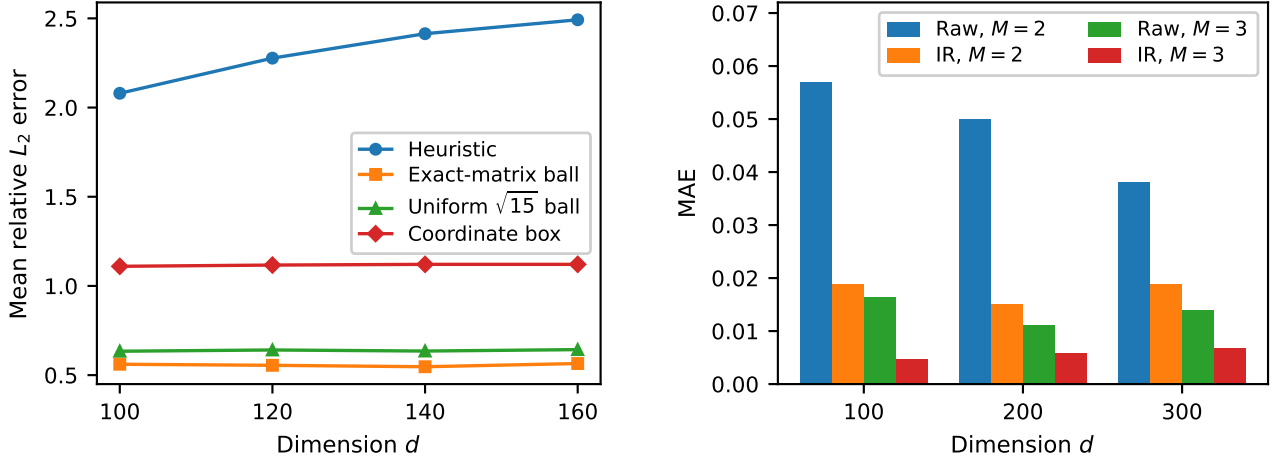
which defines a state-dependent ellipsoid in normalized delta coordinates. This is a model-derived envelope; a complete viscosity-solution justification for this particular finance bound remains future work. The implementation performs a Euclidean ball projection after delta normalization, equivalently a weighted-metric correction in the original gradient state.

D. Additional Experimental Results

Code, aggregate results, validation scripts, and project notes are maintained in the project repository. The counterexample implementation uses the source random-time law and matched random trees; the HJB tables use the corrected scaled Gauss–Laguerre reference rather than the superseded unscaled high-dimensional quadrature.

D.1. High-Dimensional Summary Values

Tables 3 and 4 give the values underlying Figure A3. The protocol uses 10 paired HJB repetitions over 1,000 interior and 200 boundary points, and 30 runs for the Neufeld–Wu benchmark.



(a) Rosenbrock HJB, $d = 100$ –160.

(b) Neufeld–Wu, $d = 100$ –300.

Figure A3. **High-dimensional outcomes beyond the subspace example.** (a) The uniform $\sqrt{15}$ ball and exact-matrix ball outperform the heuristic and coordinate-box controls. All HJB errors use the corrected, scaled Hopf–Cole reference. (b) The independent gradient-dependent benchmark compares raw and projected MLP at two sampling budgets over 30 runs. The panels use different metrics and should be compared within, not across, panels.

Table 3. Corrected HJB–Rosenbrock mean relative L_2 error.

d	heur.	exact ball	$\sqrt{15}$ ball	box
100	2.080	0.561	0.634	1.110
120	2.277	0.555	0.641	1.117
140	2.414	0.547	0.635	1.121
160	2.492	0.565	0.643	1.121

Table 4. Neufeld–Wu benchmark, 30 runs.

d	M	raw MAE	IR MAE	reduction
100	2	0.05687	0.01888	66.8%
100	3	0.01637	0.00469	71.4%
200	2	0.04989	0.01505	69.8%
200	3	0.01103	0.00580	47.5%
300	2	0.03813	0.01887	50.5%
300	3	0.01397	0.00668	52.2%

D.2. Earlier HJB Mechanism Diagnostics

Figure A4 preserves the original two-method generator-MSE and nonlinear-correction-variance plots. These values come from the earlier experiment, whereas Figure 4 compares all three methods within one newly paired experiment. No Batch-IR series is spliced into the old curves.

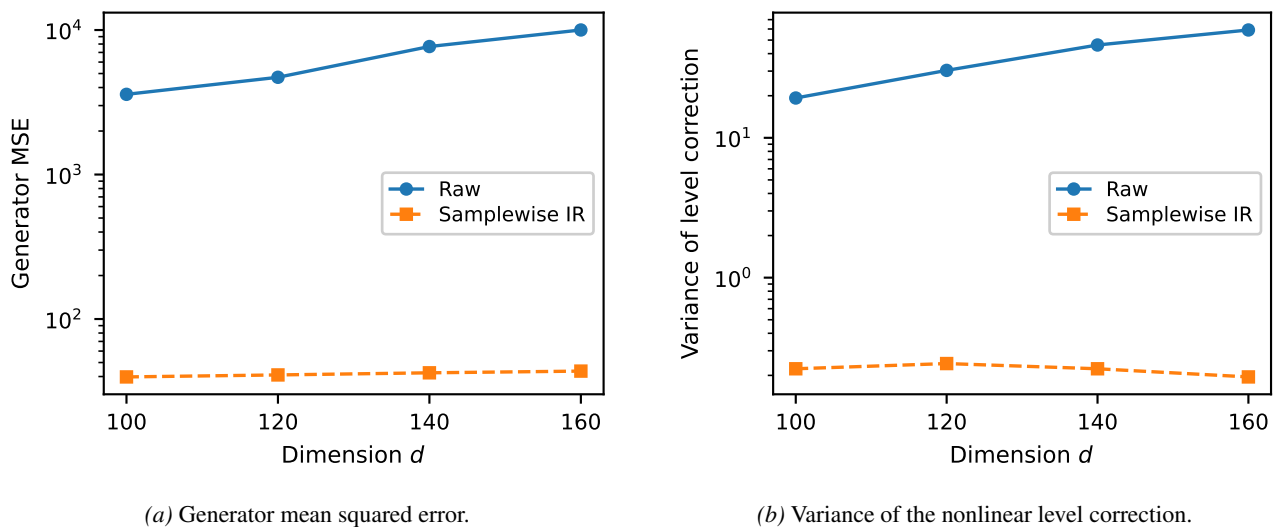


Figure A4. **The mechanism is visible inside the HJB recursion.** Using the non-oracle $\sqrt{15}$ ball, (a) generator MSE against a stable reference gradient falls by 98.9–99.6%; (b) the variance across paired runs of the actual Monte Carlo level correction falls by 98.8–99.7%. The second panel is not the variance of a single gradient coordinate or only of an individual generator sample.

D.3. Funding Error Versus Work

Figure A5 preserves the earlier full funding budget curve rather than showing only two representative settings. The vertical axis is the best observed MAE at or below the given budget, so the curve is a cumulative budget envelope, not the error of a single fixed-depth solver at every horizontal coordinate. The horizontal axis counts nonlinear-generator evaluations, not end-to-end wall time.

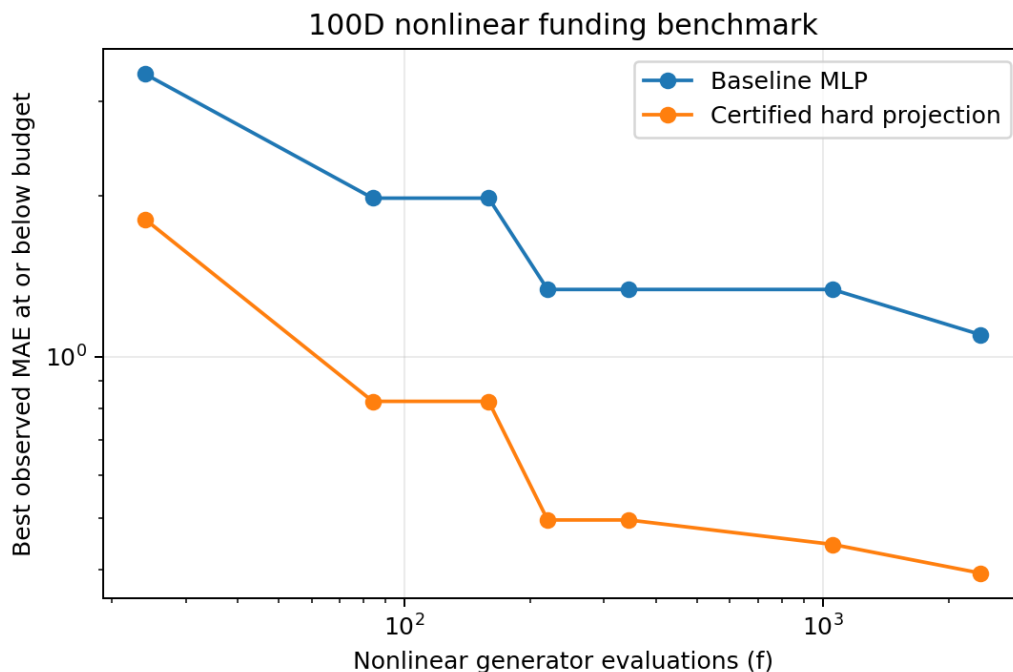
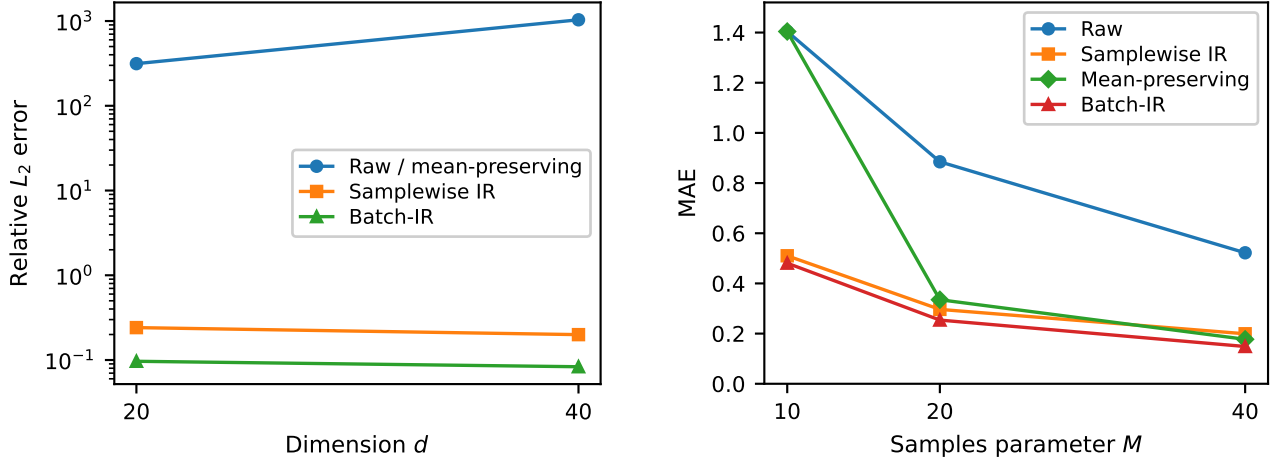


Figure A5. **Archived 100D funding error–work curve.** The entire recorded budget envelope is retained, including intermediate settings omitted from the concise main-text examples. Results are shown against nonlinear-generator evaluation count; this is not a wall-clock speedup claim.

D.4. The Batchwise Tradeoff in Numbers

The comparison in Figure A6 is summarized in Table 5. The mean-preserving diagnostic coincides with raw output when its feasibility condition fails; the funding plot additionally shows its nontrivial behavior at larger budgets.



(a) Quadratic HJB, $n = 2$, $M = 6$.

(b) 100D nonlinear funding.

Figure A6. **Two feasible retractions, different finite-budget behavior.** Corrections precede the generator; the final root value is not clipped. (a) The mean-preserving diagnostic coincides with raw MLP when the raw batch mean is infeasible. (b) Its benefit becomes visible at larger funding budgets; Batch-IR is strongest throughout this paired sweep. The HJB panel reports relative L_2 error, while the funding panel reports MAE. The panels show a finite-budget tradeoff, not universal dominance or a batchwise consistency theorem.

Table 5. Paired Samplewise IR vs. Batch-IR. Lower is better.

problem	setting	raw	sample	batch
HJB	$d = 20, M = 6$	314.136	0.2412	0.0966
HJB	$d = 40, M = 6$	1034.262	0.1993	0.0833
Funding	$M = 10$	1.4042	0.5102	0.4807
Funding	$M = 20$	0.8848	0.2965	0.2539
Funding	$M = 40$	0.5221	0.1993	0.1486

D.5. Constraint Activity and Negative Controls

Figure A7 retains the archived first-level HJB violation-rate diagnostic. It measures constraint activity before correction, which does not require the subsequently corrected numerical reference solution. In contrast, the older high-dimensional error curves based on unscaled reference quadrature are superseded by Figure A3(a).

The separate 36-setting overshoot diagnostic is now reproduced in Figure 5(a), with its companion violation-frequency plot and complete protocol in Appendix F. The fresh run gives descriptive correlations 0.886 and 0.995, compared with 0.889 and 0.996 in the older corrected-reference report. These are different Monte Carlo draws; neither series is reconstructed from image coordinates.

For the Allen–Cahn control, $d = 100$, $n = 3$, and 100 paired seeds are used. The $[0, 1]$ invariant interval is never violated at the tested budgets. Figure A8 therefore shows overlapping error curves, not merely similar averages: the recorded baseline and IR outputs are identical because the projection never activates.

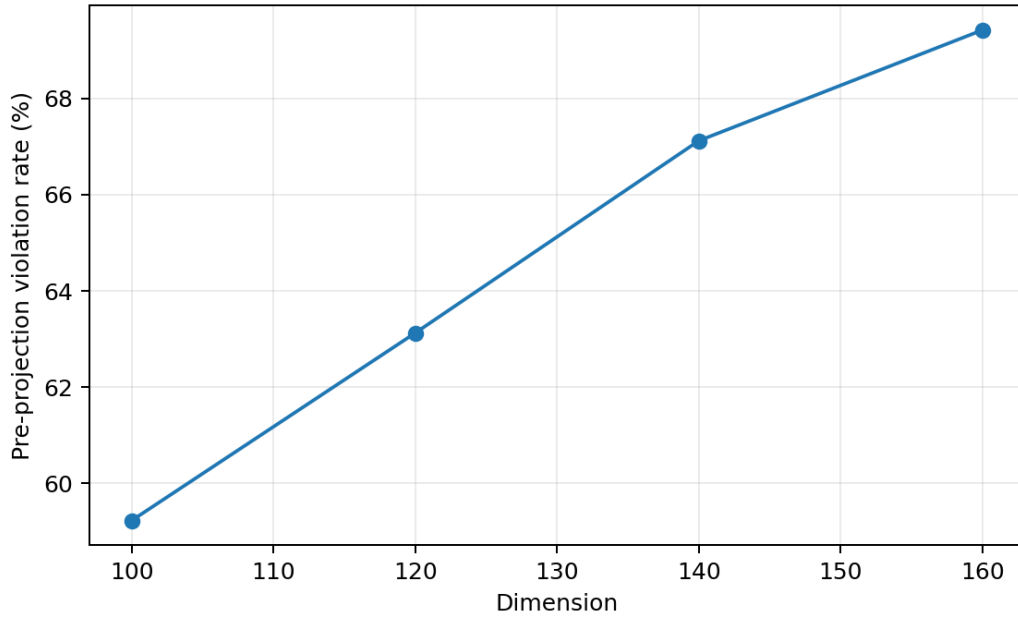


Figure A7. **The HJB constraint is active at finite budget.** The archived pre-projection violation rate increases across the tested dimensions. This is a state-feasibility diagnostic, not a reference-solution error curve and not the separate overshoot-energy/radius sweep.

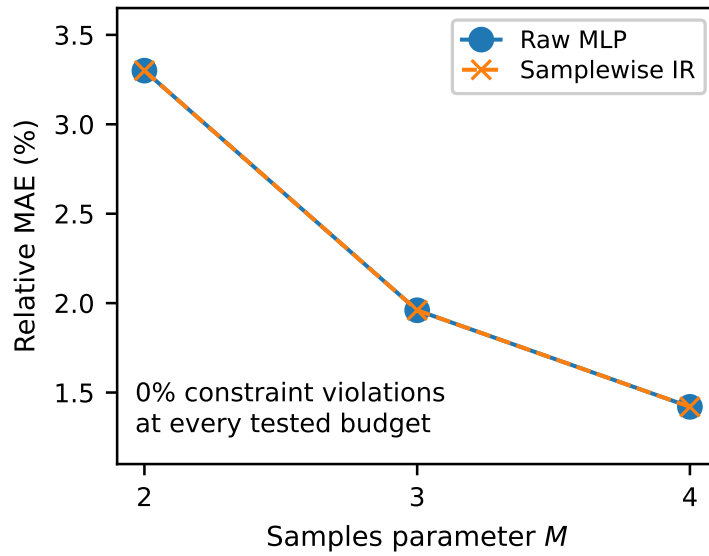


Figure A8. **An inactive constraint is a negative control.** Relative MAE is 3.30%, 1.96%, and 1.42% at $M = 2, 3, 4$ for both methods, with zero recorded violations. This figure is re-typeset from the archived summary values, not from a new experiment. Distinct markers expose the exact overlap.

E. Extended Batchwise Experiments

This appendix integrates the archived extended Batch-IR suite. It does not replace the preceding experiments or pool different runs into one comparison. The companion source package records the repository revision, source paths, and extracted numbers in `data/batchwise_extended/PROVENANCE.json`. HJB means, sample standard deviations, paired differences, win counts, and paired tests are independently checked from the archived repetition-level values; funding uncertainty is limited to what is recorded in its summary.

E.1. Headline HJB: Protocol and Paired Results

The script `hjb_batchwise_headline.py` uses

$$u_t + \Delta u - \|\nabla u\|_2^2 = 0, \quad g(x) = \log\left(\frac{1 + x^\top A x}{2}\right), \quad T = 1,$$

with the positive semidefinite Rosenbrock matrix described in Appendix C. Each dimension has a fixed matrix and a fixed test set of 1,000 points sampled inside the unit ball and 200 on the unit sphere. Every point is assigned a time sampled uniformly in $[0, 1]$. The experiment uses $n = 2$, $M = 10$, ten paired repetitions, and the same certified radius $\sqrt{15}$ in all four dimensions. No neural surrogate is trained or evaluated.

For a parent (t, x) , the value uses M^2 terminal samples and an M -sample nonlinear correction. Each child gradient uses M centered terminal samples. For $h = T - s$, $X_T = X_s + \sqrt{2h}G$, the child estimator is

$$\hat{z}_1(s, X_s) = \frac{1}{M} \sum_{k=1}^M [g(X_T^{(k)}) - g(X_s)] \frac{G_k}{\sqrt{h}}, \quad z = \sqrt{2} \nabla u.$$

Sibling child gradients are projected independently or rescaled by one common factor for each parent. Only z is corrected; the root value and terminal blocks are unchanged. The HJB implementation samples the child time uniformly. Its centered terminal differences regularize these particular gradient estimates; this protocol is not a substitution of uniform times into the general gradient-dependent MLP convergence theorem. The implementation floors time intervals inside square roots at 10^{-14} for numerical arithmetic.

The value and gradient references use 32-node Gauss–Laguerre quadrature with the scaled variable

$$c = 1 + x^\top A x + 2(T - t) \operatorname{tr}(A), \quad r = cs,$$

and log-sum-exp evaluation. These are numerical references, not exact closed-form values. The same references and test points are used by all three methods. Each repetition supplies one relative L_2 error over the entire 1,200-point set; the 1,200 points are not treated as 1,200 independent repetitions.

Table 6. Headline HJB: mean relative L_2 value error $\pm$ sample SD over ten paired repetitions. Reduction is relative to Samplewise IR. The raw method has no correction.

d	Raw	Samplewise IR	Batch-IR	Reduction	Wins	Paired p
100	2.27995 ± 0.03368	0.62077 ± 0.00222	0.26530 ± 0.00265	57.3%	10/10	4.90×10^{-20}
120	2.66715 ± 0.03591	0.62851 ± 0.00266	0.25586 ± 0.00371	59.3%	10/10	8.78×10^{-19}
140	3.09462 ± 0.06041	0.62791 ± 0.00231	0.24556 ± 0.00287	60.9%	10/10	3.79×10^{-20}
160	3.47939 ± 0.03308	0.64044 ± 0.00343	0.24287 ± 0.00239	62.1%	10/10	1.94×10^{-19}

All four 10/10 win counts are within-dimension comparisons on fixed matrices and test sets, not forty independent problem instances. Table 6 reports descriptive two-sided paired t -tests without multiple-comparison adjustment. These tests concern the sampled repetitions of this fixed benchmark, not generalization over PDE instances. In particular, the new *Raw* column means no correction at all; the earlier *heuristic* column in Table 3 uses clipping and must remain a separate baseline. The different raw and corrected values across old and new studies are not paired observations.

E.2. Matched Generator and Correction Diagnostics

For each repetition, the script retains complete sibling groups for the first 64 parent points. Ten children per parent and ten repetitions yield 6,400 diagnostic child states. The retained parent subset consists of interior test points. For a method a ,

let $\tilde{z}_{rji}^a$ denote its corrected child gradient at the sampled point S_{rji} , and let $z_{\text{ref}}(S_{rji})$ be the quadrature reference. The generator diagnostic is

$$\widehat{\text{MSE}}_f^a = \frac{1}{10 \cdot 64 \cdot 10} \sum_{r=1}^{10} \sum_{j=1}^{64} \sum_{i=1}^{10} [f(\tilde{z}_{rji}^a) - f(z_{\text{ref}}(S_{rji}))]^2.$$

The outer correction is evaluated at *all* $J = 1200$ fixed parent points:

$$C_{rj}^a = \frac{T - t_j}{M} \sum_{i=1}^M f(\tilde{z}_{rji}^a), \quad \bar{C}_j^a = \frac{1}{R} \sum_{r=1}^R C_{rj}^a,$$

$$\hat{V}_{\text{corr}}^a = \frac{1}{J} \sum_{j=1}^J \frac{1}{R-1} \sum_{r=1}^R (C_{rj}^a - \bar{C}_j^a)^2, \quad R = 10.$$

This is the across-repetition variance of the actual averaged nonlinear correction, followed by an average over test points. It is neither the variance of a single gradient coordinate nor the variance of the entire root value.

Table 7. Matched HJB mechanism diagnostics. Generator MSE and averaged-correction variance use the definitions and parent sets given in the text.

d	Generator MSE			Averaged-correction variance		
	Raw	Samplewise IR	Batch-IR	Raw	Samplewise IR	Batch-IR
100	4427.03	38.42	10.50	20.1419	0.2221	0.1403
120	9992.13	43.17	10.98	29.9815	0.2156	0.1393
140	12730.85	42.62	10.81	45.2710	0.2058	0.1409
160	11312.72	42.99	10.30	60.7858	0.2109	0.1526

For comparison, the script also computes the sample variance of the individual $f(\tilde{z})$ values pooled across the 6,400 diagnostic states. That statistic mixes variation across parent locations, child times, sibling draws, and repetitions. It should not be interpreted as a pointwise Monte Carlo variance or as the variance of an average with independent terms. Table 8 deliberately retains the unfavorable cases: Batch-IR has larger pooled individual-increment variance than Samplewise IR in three dimensions, despite lower averaged-correction variance in all four. The two statistics also use different parent sets.

Table 8. Pooled individual-increment variance on the diagnostic child states. Positive change means Batch-IR is worse on this particular statistic; no uniform dominance is claimed.

d	Raw	Samplewise IR	Batch-IR	Change vs. sample
100	3384.7226	6.1441	5.7609	-6.2%
120	7370.8108	4.8560	5.8406	+20.3%
140	9908.3877	5.1520	5.8845	+14.2%
160	8811.4296	4.8966	5.8071	+18.6%

The common factor is active in almost every headline HJB sibling batch. Table 9 reports the recorded rate of $\alpha_B < 1$ and the mean factor, aggregated over 1,200 parent batches per repetition. The lower final errors therefore do not demonstrate that the retraction becomes negligible. We do not infer a bias estimate, a vanishing-correction limit, or general Batch-IR consistency from these activity statistics.

Table 9. Common-scale activity in the headline HJB study. These are sibling-batch statistics, not scalar-coordinate violation rates.

d	Batch activation	Mean α_B
100	99.867%	0.3297
120	99.967%	0.3013
140	99.950%	0.2744
160	99.983%	0.2598

E.3. Deeper 100D Funding Recursions

The script `funding_batchwise_deep.py` uses $d = 100$, $T = 0.5$, volatility $\sigma = 0.2$, drift $\mu = 0.06$, lending rate $R_l = 0.04$, borrowing rate $R_b = 0.06$, and initial asset prices $x_i = 100$. Its payoff and driver are

$$g(x) = (\max_i x_i - 120)_+ - 2(\max_i x_i - 150)_+,$$

$$f(y, z) = -R_l y - \frac{\mu - R_l}{\sigma} \sum_i z_i + (R_b - R_l) \left(\frac{\sum_i z_i}{\sigma} - y \right)_+.$$

The geometric-Brownian increments are simulated directly. Random times use $r \sim \text{Beta}(1/2, 1)$ with inverse-density weights; the high and low branches use separate child random draws while the three methods share those draws. Each completed child is normalized to delta coordinates $\Delta_i = z_i / (\sigma x_i)$. For siblings at distinct times s_i , the batch factor is

$$\alpha_B = \min \left\{ 1, \min_i \frac{\exp(\sigma^2(T - s_i)/2)}{\|\hat{\Delta}_i\|_2} \right\},$$

with the usual no-shrinkage convention for zero norm. The implementation then maps back to z and evaluates the driver without changing y . This uses the model-derived envelope whose remaining analytic justification is stated in Appendix C; the new experiments do not upgrade it into a new certificate theorem.

Each (n, M) configuration uses 100 paired root estimates, computed in blocks of ten. A block is a computational grouping of roots, not the sibling group used to calculate α_B . The value reference is the recorded numerical value 21.299. All reported MAEs and pairwise win fractions are measured against that reference.

Table 10. Extended 100D funding: 100 paired root estimates per setting; all MAEs use reference 21.299. Paired tests are reported by the archived summary; no replica-level confidence intervals are inferred.

n	M	Raw MAE	Samplewise IR	Batch-IR	Reduction	Wins	Reported p
2	10	1.52683	0.52147	0.45658	12.4%	73/100	4.73×10^{-6}
3	8	1.10714	0.38394	0.30012	21.8%	75/100	3.18×10^{-10}
4	3	3.20564	0.77079	0.68970	10.5%	60/100	1.42×10^{-3}

The smaller Batch-IR MAE survives at $n = 3$ and $n = 4$. Because M changes across rows, the table is not evidence that raising depth alone reduces error. The imported summary records a 100% batch activation rate at all three settings and mean common factors of 0.1632, 0.1748, and 0.1218. The paired p -values in Table 10 are carried from the repository summary. The imported funding results do not include the per-replica arrays needed to recompute confidence intervals or those tests, so no uncertainty bars have been reconstructed from aggregate MAEs.

E.4. Neufeld–Wu at the Smallest Extended Budget

The script `neufeld_batch_m2.py` uses $n = M = 2, 30$ repetitions at each of $d = 100, 200, 300$, linear drift $0.06x$, additive diffusion $0.2I$, $T = 0.25$, and the max-asset call-spread payoff with strikes 95 and 120. The driver is $(10/d)(\|z\|_\infty - 25)_+$, where this implementation’s gradient coordinate is ∇u , rather than $\sigma^\top \nabla u$. It uses a 12-step Euler forward approximation and $\text{Beta}(1/2, 1/2)$ random times. The reference is numerical integration of the maximum-of-Gaussians payoff under the continuous linear diffusion; consequently the errors include the implementation’s forward/time-discretization effects. They are not exact-time MLP asymptotics.

Table 11. Neufeld–Wu extension at $n = M = 2, 30$ repetitions per dimension. Samplewise IR and Batch-IR tie; this table must not be pooled with the older runs in Table 4.

d	Reference	Raw MAE	Samplewise IR MAE	Batch-IR MAE
100	6.76395833	0.06066	0.02131	0.02131
200	6.78798317	0.03878	0.01682	0.01682
300	6.80125502	0.04039	0.01737	0.01737

Both corrections map the offending gradients well below the threshold 25, so the same spurious nonlinear source is removed. The archived summary gives identical Samplewise/Batch-IR MAEs, with zero reported discrepancy; these ties must not be counted as wins or losses. Its recorded 20% activation rate includes calls on trivial zero-depth child states in the denominator and is not the violation rate among nontrivial gradients. The earlier $n = M = 3$ batch study likewise ties, with corrected MAEs 0.00665, 0.00711, and 0.00679. It uses a different budget and sample realization, so the two studies are reported separately rather than pooled.

E.5. Batchwise Structural Sanity Check

The script `counterexample_batchwise_sanity.py` uses $n = m = 3$, 512 repetitions, and $d = 10, 100, 1000$. Within each sibling group it first applies the certified Q_d and then uses

$$\alpha_B = \min\{1, 1/\max_i |(Q_d z_i)_1|\}, \quad \tilde{z}_i = \alpha_B Q_d z_i.$$

Since f_d vanishes on S_d , the nonlinear terms are zero even when $\alpha_B < 1$. Random time and active-coordinate streams are shared across dimensions; inactive-coordinate randomness is separate. The observed corrected value RMSE is consequently identical across the dimensions, as expected from this coupling.

Table 12. Separate batchwise structural sanity check: value RMSE over 512 paired replicas at $n = m = 3$. Maximum discrepancy compares paired values, not gradients or full states.

d	Raw	Subspace IR	Batch structural	Max. paired discrepancy
10	3.595016	0.110083	0.110083	0
100	39.121427	0.110083	0.110083	0
1000	396.591466	0.110083	0.110083	0

The recorded activation rate is 0.58984375 and mean common factor is 0.77601904 at all three dimensions. The maximum paired value discrepancy is zero. This is a value-output identity check in a separate run, not a replacement for the primary counterexample sweep or independent Gaussian-moment test in Figures 2 and A1.

Root-gradient distinction. The code projects returned gradients in the samplewise subspace mode, but applies $\alpha_B Q_d$ only inside the batch mode’s generator calls. It does not project the batch mode’s final returned gradient. Thus its exact value equality cannot establish equal full-state MSE or the constant $(4 - 2/\pi)/N$ for that unprojected batch output. The structural part of Theorem 2 is what this sanity check tests. Pure common radial scaling without Q_d is not an exact-rescue method.

F. Overshoot Energy, Constraint Activity, and Final Gain

This section supplies the complete protocol and data behind the 36-setting mechanism sweep in Figure 5(a). It is an independent reproduction of the earlier overshoot-energy diagnostic, not a claim to recover the original Monte Carlo draws. The earlier corrected-reference report gave Spearman correlations 0.889 (energy versus gain) and 0.996 (violation rate versus gain). The present run gives 0.886 and 0.995, respectively. The two runs are not pooled, and the older unscaled-reference sweep is not used here.

F.1. Fixed Depth, Paired Budgets, and Valid Radii

We use the Rosenbrock HJB from Appendix C, with $d = 100$, $T = 1$, $n = 2$, and $z = \sqrt{2}\nabla u$. The terminal datum is $g(x) = \log((1 + x^\top Ax)/2)$. The coefficient vectors defining A are sampled independently from $[0.5, 1.5]$ with NumPy seed 0. Its exact-matrix gradient certificate is

$$R_0 = \sqrt{2\lambda_{\max}(A)} = 3.556491956352001.$$

The six radii aR_0 , for $a \in \{1, 1.25, 1.5, 2, 3, 4\}$, all contain the exact gradient. We cross them with $M \in \{2, 3, 4, 6, 8, 10\}$, keeping depth fixed. For each budget, the same raw stochastic tree is reused across all six radii and the unprojected baseline.

A fixed test set has 1,000 interior unit-ball points, 200 unit-sphere points, and independent times uniform in $[0, 0.95]$, sampled with seed 2126. Ten independent repetitions are evaluated in chunks of 32 points, with seed sequences [98731, 100, rep, chunk start]. All arithmetic is float64. The root terminal block uses M^2 draws; each of M randomly timed

children uses M centered terminal draws for its gradient. Only the *completed child gradient* entering $f(z) = -\|z\|^2/2$ is projected. Neither the root value nor a terminal sample is clipped.

The child time is uniform between the parent time and T . This is a fixed-depth *value-channel* experiment; it is not a convergence claim for a general uncentered gradient-weighted recursion with uniform time. No time-denominator floor is used. The reference is scaled Gauss–Laguerre Hopf–Cole quadrature with 64 nodes. A 128-node cross-check differs by at most 1.8×10^{-15} on the evaluation set; this is quadrature agreement, not a universal certified error bound.

F.2. What Each Axis Measures

Let $J = 1200$, $K = 10$, and let $\hat{z}_{rji}$ be child i at parent point j in repetition r . For radius $R = aR_0$ we record

$$\hat{\Omega}_{M,a} = \frac{1}{KJM} \sum_{r,j,i} (\|\hat{z}_{rji}\| - R)_+^2, \quad \hat{p}_{M,a} = \frac{1}{KJM} \sum_{r,j,i} \mathbf{1}\{\|\hat{z}_{rji}\| > R\}.$$

Each repetition gives a relative ℓ_2 value error on the entire fixed point set. Write e_M^{raw} and $e_{M,a}^{\text{IR}}$ for their averages over the ten repetitions. The plotted gain is

$$G_{M,a} = 1 - \frac{e_{M,a}^{\text{IR}}}{e_M^{\text{raw}}}.$$

Thus each main-figure curve holds M fixed and varies only the valid radius; the companion Figure A9 replaces overshoot energy by violation frequency. The six budgets and six radii share data, and the correlations are descriptive. They do not establish that increasing violations while holding all other quantities fixed causes an improvement.

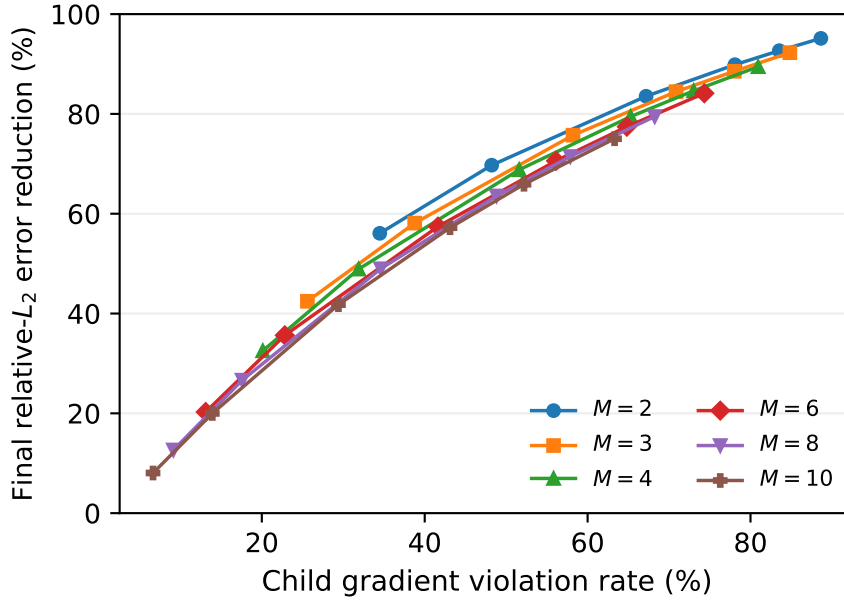


Figure A9. Violation frequency complements overshoot energy. The same newly executed 36 settings as Figure 5(a) are shown. Each line fixes M and varies the certified radius from $1\times$ to $4\times$. Larger valid radii reduce the frequency of correction. Across the full shared-data sweep, the descriptive Spearman correlation is 0.995.

All value arrays, evaluation points, the matrix, references, per-repetition energies and rates, and summarized rows are included with the reproduction script. No coordinates were digitized from a figure.

Table 13. All 36 settings of the independent, corrected-reference reproduction. a multiplies the exact-matrix certificate R_0 ; p is the pre-projection violation fraction. Raw and IR columns are mean relative L_2 errors over ten paired repetitions. These are newly sampled results, not reconstructed old figure coordinates.

M	a	$\widehat{\Omega}$	p (%)	Raw	IR	Gain (%)
2	1	205.627	88.64	14.7731	0.7180	95.14
2	1.25	188.722	83.54	14.7731	1.0826	92.67
2	1.5	173.139	78.09	14.7731	1.4973	89.86
2	2	145.591	67.18	14.7731	2.4311	83.54
2	3	102.620	48.22	14.7731	4.4727	69.72
2	4	71.954	34.48	14.7731	6.4891	56.07
3	1	127.685	84.83	8.9572	0.6925	92.27
3	1.25	114.912	78.01	8.9572	1.0260	88.55
3	1.5	103.372	70.86	8.9572	1.3918	84.46
3	2	83.548	58.21	8.9572	2.1738	75.73
3	3	54.292	38.77	8.9572	3.7509	58.12
3	4	34.961	25.60	8.9572	5.1545	42.45
4	1	88.555	80.94	6.3795	0.6722	89.46
4	1.25	78.243	73.02	6.3795	0.9806	84.63
4	1.5	69.086	65.28	6.3795	1.3105	79.46
4	2	53.756	51.60	6.3795	1.9895	68.81
4	3	32.246	31.89	6.3795	3.2594	48.91
4	4	18.991	20.12	6.3795	4.3045	32.53
6	1	52.753	74.33	3.9715	0.6299	84.14
6	1.25	45.298	64.82	3.9715	0.8959	77.44
6	1.5	38.870	56.12	3.9715	1.1678	70.59
6	2	28.553	41.63	3.9715	1.6907	57.43
6	3	15.189	22.82	3.9715	2.5561	35.64
6	4	7.810	13.11	3.9715	3.1664	20.27
8	1	35.913	68.24	2.8884	0.5929	79.47
8	1.25	30.094	57.88	2.8884	0.8243	71.46
8	1.5	25.191	48.81	2.8884	1.0526	63.56
8	2	17.583	34.60	2.8884	1.4739	48.97
8	3	8.355	17.59	2.8884	2.1187	26.65
8	4	3.757	9.14	2.8884	2.5227	12.66
10	1	26.444	63.31	2.2462	0.5609	75.03
10	1.25	21.696	52.19	2.2462	0.7658	65.91
10	1.5	17.776	43.10	2.2462	0.9614	57.20
10	2	11.867	29.41	2.2462	1.3078	41.78
10	3	5.106	13.92	2.2462	1.7975	19.98
10	4	2.032	6.64	2.2462	2.0655	8.04

G. Additional Positive Benchmarks and Geometry Ablations

The older benchmark records answer questions not replaced by the three-method Batch-IR extension. The corrected HJB comparison in Figure A3(a) tests certified joint geometry against coordinate clipping and heuristic clipping. The two-budget Neufeld–Wu curves in Figure 5(b) and Table 4 test an independent gradient-dependent implementation. Their archived 30-run records have been recovered; all six dimension–budget pairs, including the coordinate-box ties, are retained. The box and joint projection agree in value in this thresholded problem, so these results cannot establish that a ball is superior to a box for every generator.

G.1. The Original Funding Work Curve and Higher-Level Geometry

Figure A5 retains the original budget-envelope image. Table 14 gives every recorded setting behind it, and Figure A10(a) plots all settings without replacing them by a running minimum. In particular, the projected method has MAE 0.4953 at 220 generator calls; the raw method still has MAE 1.0996 at 2,384 calls in this recorded sweep. These are benchmark-specific observations, not a cost-to-accuracy theorem or a wall-clock speedup.

The separate higher-level ablation in Figure A10(b) reaches depth five and keeps the coordinate-box control. At $(n, M) = (4, 3)$, the raw, box, and joint-IR MAEs are 3.0252, 2.7077, and 0.8772; at $(5, 2)$ they are 5.1956, 4.5922, and 1.4590. Hence the positive evidence is not restricted to the shallow root or to a comparison against an entirely unprotected method. The methods are paired within each row. The three settings have different budgets and replication counts; a depth-only trend is not inferred.

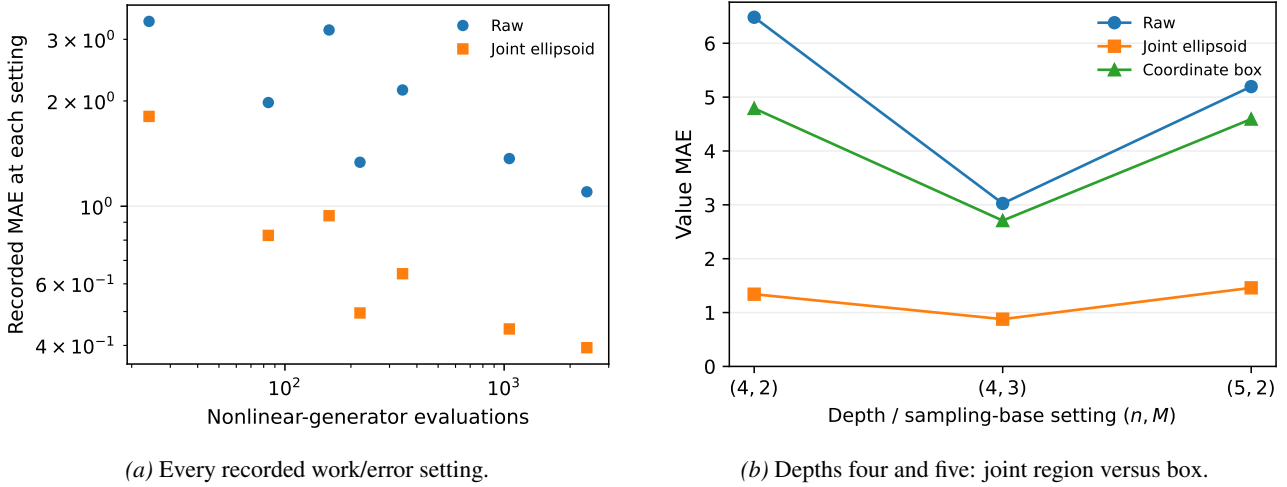


Figure A10. **Funding evidence not subsumed by the new Batch-IR comparison.** (a) Dots show all seven original settings; the original cumulative envelope is separately preserved in Figure A5. (b) A paired older geometry ablation includes the coordinate box and a fifth-level calculation. No Batch-IR series is spliced into these earlier experiments. The connecting lines in (b) only organize settings, not a continuous work or depth axis.

Table 14. All recorded settings behind the original funding work envelope. Counts are nonlinear-generator evaluations; replication counts differ across settings.

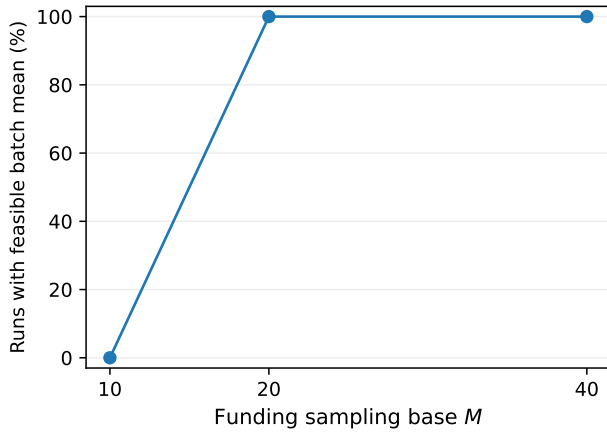
n	M	Runs	f calls	Raw MAE	Joint IR MAE
2	3	100	24	3.3759	1.8063
2	6	100	84	1.9789	0.8253
2	10	100	220	1.3357	0.4953
3	3	100	159	3.1904	0.9393
3	4	100	344	2.1495	0.6413
3	6	60	1056	1.3686	0.4459
3	8	40	2384	1.0996	0.3941

Table 15. Earlier higher-level funding geometry ablation. The three methods are paired within each setting; no Batch-IR series is inferred from these data.

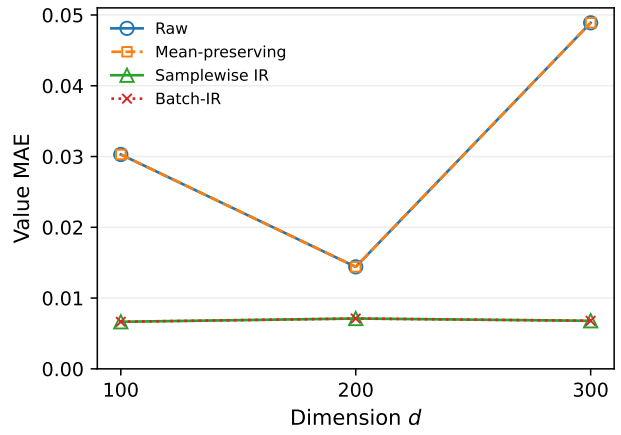
n	M	Runs	f calls	Raw	Joint IR	Box
4	2	60	252	6.4827	1.3411	4.7908
4	3	30	1032	3.0252	0.8772	2.7077
5	2	30	1124	5.1956	1.4590	4.5922

G.2. Mean Feasibility and the Earlier Thresholded-Driver Comparison

The earlier batch study includes two further diagnostics. In the funding sweep, the mean-preserving rule is feasible in 0%, 100%, and 100% of recorded runs at $M = 10, 20, 40$. This concerns that diagnostic’s feasible-mean condition, not the fraction of individual children inside the set. At $n = M = 3$ in Neufeld–Wu, the mean-preserving rule coincides with raw MLP, whereas Samplewise IR and Batch-IR have identical lower MAEs. These are separate results from the $n = M = 2$ extension in Appendix E.4.



(a) Funding: feasible mean for the diagnostic.



(b) Neufeld–Wu: the earlier $n = M = 3$ batch study.

Figure A11. **Feasibility and ties are part of the evidence.** Reported aggregate values are re-plotted without invented replication-level uncertainty. In (b), distinct markers expose two exact overlaps: Raw with the mean-preserving diagnostic, and Samplewise IR with Batch-IR. Preventing the same threshold activation gives a tie, not a Batch-IR win.

H. Controlled Surrogate-Quality Diagnostic

An archived 20D logistic test probes total-state constraints in a defect-correction setting. We retain it as a *controlled diagnostic*, not as a comparison against trained SCaSML checkpoints. The surrogate is defined from the analytic solution as $\hat{u} = qu^*$, with $q \in \{1, 0.95, 0.75, 0.5, 0\}$. Its relative error is $1 - q$. At depth $n = 2$, three sampling bases $M \in \{1, 2, 4\}$ are tested with 40 repetitions for each setting. The methods are raw defect MLP, fixed defect clipping to $[-0.01, 0.01]$, and correction using the total-state bounds $0 \leq u \leq 1, 0 \leq z_i \leq 1/16$ recorded for this logistic test.

The noisiest budget illustrates why a fixed small defect range is not robust to surrogate quality. With no surrogate ($q = 0$, $M = 1$), raw and total-state-IR relative errors are 19.87% and 18.37%, whereas fixed defect clipping gives 98.44%. The true correction is no longer small enough for the chosen defect clip. This does not establish that total-state IR always helps: at $M = 4$ it is worse than raw defect MLP at every nonperfect surrogate setting in the archive. For example, at 25% surrogate error the errors are 1.357% (raw) and 1.714% (IR). All such rows are included in Table 16.

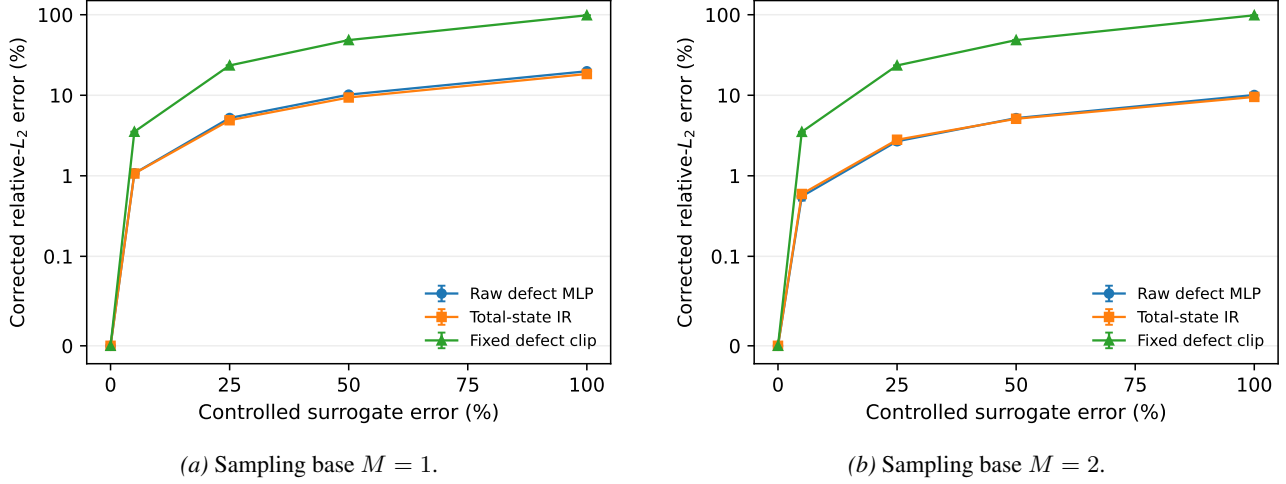


Figure A12. **A fixed defect clip is sensitive to surrogate quality.** The surrogate is analytically scaled, not trained. Error bars are the recorded standard deviations over 40 runs, not confidence intervals. The vertical axis is symmetric-log with a linear segment below 0.1 percentage point, retaining the exact zero-error perfect-surrogate case.

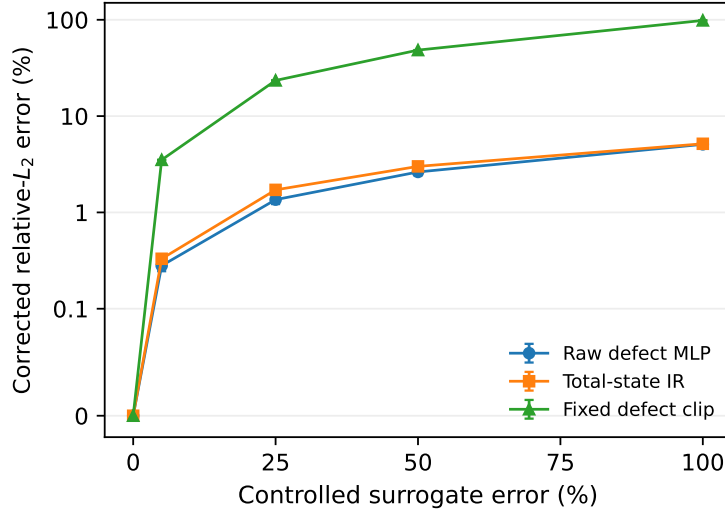


Figure A13. **The larger-budget boundary case is retained.** At $M = 4$, raw defect MLP is already sufficiently accurate that total-state correction has no general advantage and slightly worsens the nonperfect cases. This panel uses the same axis convention and uncertainty definition as Figure A12. It is not evidence against the exact counterexample theorem, which concerns a different generator and constraint.

Table 16. Controlled logistic-surrogate diagnostic: relative L_2 error in percent (mean $\pm$ recorded standard deviation over 40 runs). The perfect-surrogate rows remain, and the table also retains cases where IR is worse than raw defect MLP.

M	Surrogate error (%)	Raw defect MLP	Total-state IR	Fixed defect clip
1	0	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
1	5	1.073 ± 0.075	1.067 ± 0.062	3.524 ± 0.004
1	25	5.221 ± 0.301	4.898 ± 0.252	23.453 ± 0.001
1	50	10.178 ± 0.518	9.382 ± 0.461	48.446 ± 0.000
1	100	19.865 ± 1.037	18.368 ± 0.888	98.444 ± 0.000
2	0	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
2	5	0.555 ± 0.063	0.594 ± 0.044	3.512 ± 0.001
2	25	2.684 ± 0.234	2.804 ± 0.131	23.452 ± 0.000
2	50	5.202 ± 0.341	5.105 ± 0.238	48.446 ± 0.000
2	100	10.077 ± 0.512	9.541 ± 0.450	98.444 ± 0.000
4	0	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
4	5	0.280 ± 0.035	0.330 ± 0.022	3.510 ± 0.000
4	25	1.357 ± 0.136	1.714 ± 0.112	23.452 ± 0.000
4	50	2.630 ± 0.206	3.005 ± 0.207	48.446 ± 0.000
4	100	5.089 ± 0.346	5.159 ± 0.362	98.444 ± 0.000

The archived JSON supplies 45 method-level rows, including the perfect surrogate and all three budgets. No learned-checkpoint training cost, real-surrogate generalization result, or end-to-end SCaSML superiority is inferred from these data.

I. Batchwise Negative Controls

The repository’s paired negative-control suite complements the earlier samplewise controls. It separates two reasons for an absent value benefit: the constraint can be inactive, or it can be active on a state that is irrelevant to the generator. Figure A14 and Table 17 preserve both methods’ overlaps rather than omit zero-gain cases.

Table 17. Recorded paired negative controls. All three methods have the same error and zero maximum value discrepancy. Allen–Cahn reports relative MAE in percent; the other rows report absolute MAE. Activity columns refer to the constrained state and the Batch-IR implementation.

Problem	n	M	Common error	Violation (%)	Batch active (%)	Max Δu
Allen–Cahn	3	2	3.303465	0.00	0.00	0
Allen–Cahn	3	3	1.959025	0.00	0.00	0
Allen–Cahn	3	4	1.421529	0.00	0.00	0
Credit risk	2	10	0.450018	0.00	0.00	0
Credit risk	3	6	0.297810	0.00	0.00	0
Credit risk	4	3	0.516940	0.00	0.00	0
Linear	2	4	0.114882	100.00	100.00	0
Linear	2	8	0.050744	97.00	97.00	0
Linear	2	16	0.026660	8.50	8.50	0

I.1. Inactive Constraints: Allen–Cahn and Credit Risk

For 100D Allen–Cahn, $n = 3$ and $M = 2, 3, 4$ use the maximum-principle interval $[0, 1]$ with 100 paired repetitions. No child violates the interval, and Batch-IR uses $\alpha_B = 1$ throughout. Raw, Samplewise IR, and Batch-IR match realization by realization. The earlier Allen–Cahn panel in Figure A8 remains; the new panel explicitly includes Batch-IR.

For the 100D credit-risk experiment, the recorded parameters are $T = 2$, $\sigma = 0.2$, $\beta = 0.03$, $K_1 = 30$, $K_2 = 60$, $L = 15$, and reference value 2.626. The tested interval is $[-15, 15]$. Each setting has 100 paired repetitions, and all methods again match exactly with zero constraint activity. These values belong to the newer paired negative-control run and are not substituted for the earlier credit-risk seed set.

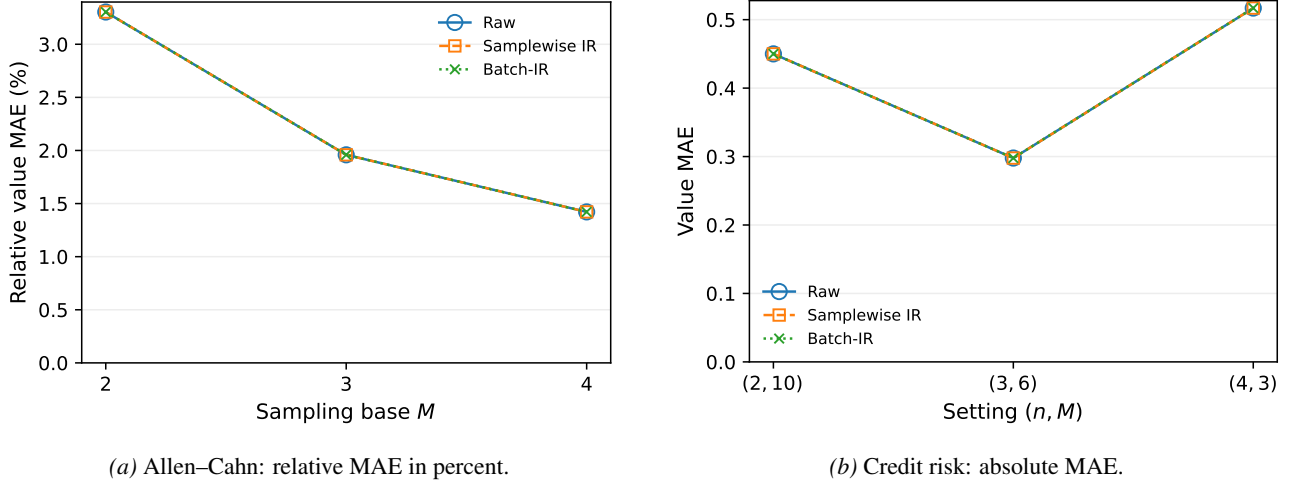


Figure A14. **Inactive constraints leave all three algorithms unchanged.** Open and crossed markers expose identical curves without horizontal jitter or artificial separation. Every recorded maximum value difference from raw MLP is zero. The credit-risk settings vary both depth and sampling base.

I.2. Active but Generator-Irrelevant Gradient Correction

For the linear 100D convection–diffusion test, $f = 0$. The noisy gradient estimate is corrected, but the value estimator does not depend on that gradient. At $M = 4, 8, 16$, the recorded gradient-violation and batch-activation rates are 100%, 97%, and 8.5%, respectively. Nevertheless the paired value outputs remain exactly identical. This is stronger than an inactive-constraint control: feasibility correction can occur frequently without improving a value that never reuses the constrained channel.

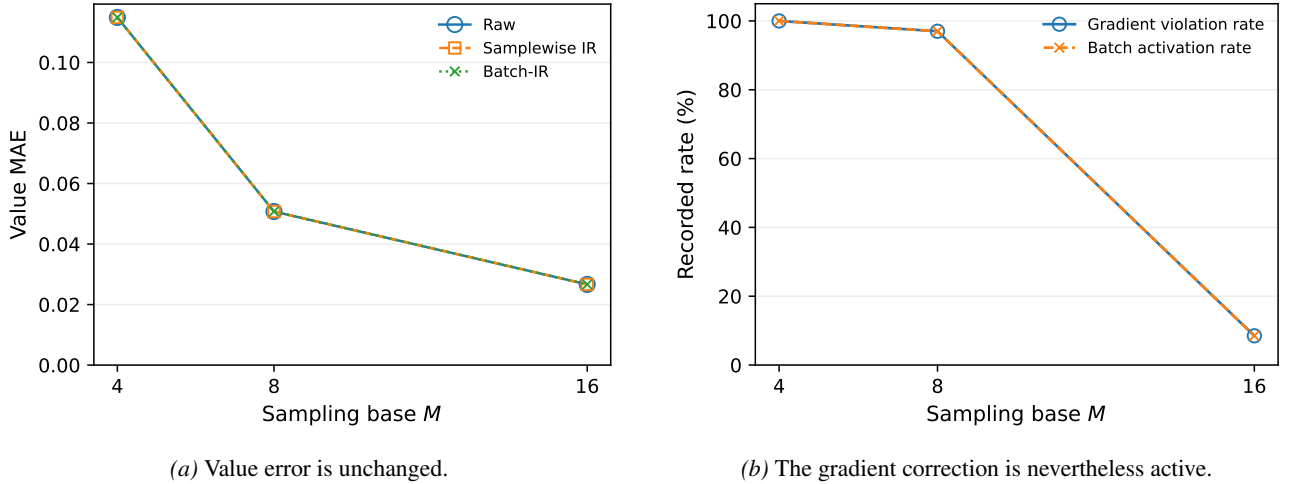


Figure A15. **Constraint activity alone is not the mechanism.** (a) All methods have equal value MAE and zero maximum paired discrepancy. (b) Gradient-violation and Batch-IR activation rates overlap but are often large. Because $f = 0$, the corrected gradient cannot affect the value path.

J. An Elliptic Neural-BSDE Diagnostic Outside the MLP Claim

For completeness, the older repository also records a diagnostic on the semilinear elliptic benchmark of [Kremsner et al. \(2020\)](#). This uses a neural random-terminal-time BSDE solver, *not* an MLP recursion, and is not counted as an additional

positive MLP benchmark. On the unit ball the equation and analytic solution are

$$\Delta u + \|\nabla u\|^2 = 2e^{-u}, \quad u|_{\partial B} = \log(2/d), \quad u(x) = \log\left(\frac{1 + \|x\|^2}{d}\right), \quad \nabla u(x) = \frac{2x}{1 + \|x\|^2}.$$

Thus $\|\nabla u\| \leq 1$ is a valid ball certificate. The run uses $d = 100$, simulated horizon 0.1, 50 time intervals, 500 Adam iterations, training batches of 64, validation size 256, and a starting point of radius 0.5.

The baseline implementation divides the gradient-network output by d . Under this scaling, the radius-one projection is inactive and the tested corrections coincide. Removing only this scaling produces a useful but mixed ablation. Across three paired seeds, the joint ball lowers the terminal squared BSDE residual from 0.0275 to 0.0103, but slightly raises the relative error of the one reported value from 1.76% to 1.80%. The coordinate box gives a smaller value error while substantially worsening the terminal loss (Table 18). Neither metric should be hidden in favor of the other.

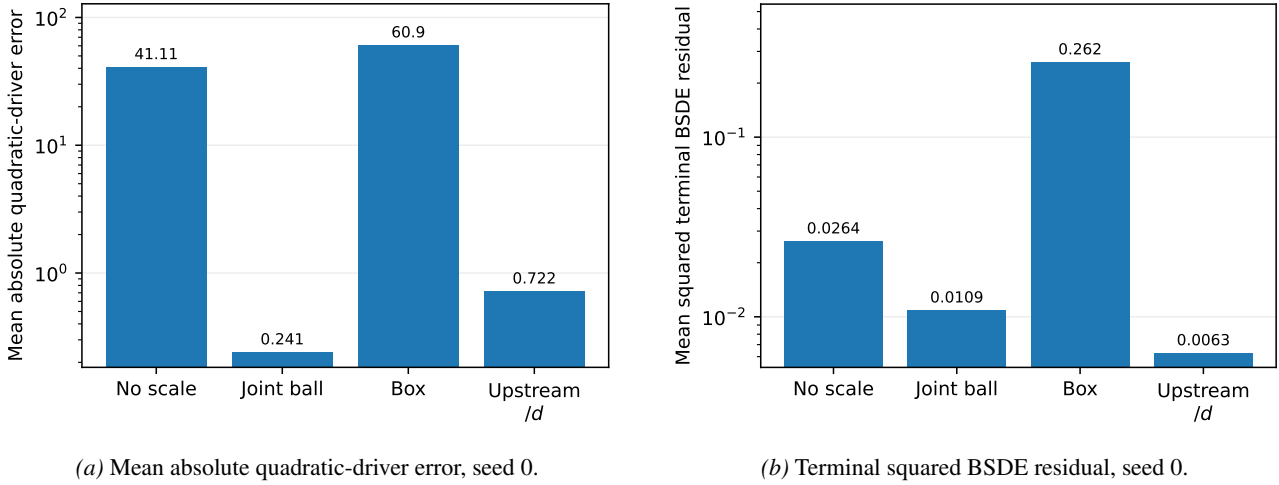


Figure A16. Different metrics can favor different corrections. These are archived seed-0 diagnostics, not averages over three seeds. The driver statistic is the mean of $\|\hat{z}\|^2 - \|\nabla u\|^2$ on active sampled states; it is not generator MSE. The upstream $/d$ rule remains best on terminal loss. This out-of-scope extension is retained as a limitation/diagnostic, not presented as an improvement over the published elliptic solver.

Table 18. Archived elliptic no-scale ablation, means over three paired seeds. Lower terminal loss does not imply lower error in the single reported value. Rounded report values are preserved. This is a neural BSDE diagnostic, not an MLP benchmark.

Method	Relative value error (%)	Terminal BSDE loss
No scale	1.76	0.0275
Joint ball	1.80	0.0103
Coordinate box	0.53	0.2682

In the archived seed-0 radius check, terminal losses are 0.0109, 0.0275, and 0.1020 at ball radii 1, 2, and 4. Relaxing the certified bound weakens this pathwise stabilization, but does not turn it into a theorem about the trained solver or the reported value error. These figures use the rounded source-report values; no new neural training was performed for this manuscript update.